MovieLens Project Report

Predictive Modeling for Movie Recommendations

A REPORT PRESENTED
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Contents

1 Executive Summary

This report presents the development of a high-performance recommendation engine based on the **MovieLens 10M dataset**. The project was conducted as part of the **HarvardX Data Science Professional Certificate** capstone requirements.

Problem Statement: The primary challenge was to predict user movie ratings with high precision while managing a massive dataset (10 million observations) under strict computational constraints (6 GB of RAM).

Methodology: Our approach followed a rigorous data science workflow:

1. **Exploratory Data Analysis (EDA):** Identification of systemic biases related to specific movies, individual user behavior, and temporal drifts.
2. **Feature Engineering:** Extraction of movie age and weekly temporal bins to capture non-linear rating trends.
3. **Modeling:** We developed an incremental linear model, moving from a simple global average to a multi-factor architecture. To handle the "Cold Start" problem and ensure statistical reliability, we implemented **Regularization**.
4. **Computational Strategy:** To respect memory limits, we utilized an iterative **Backfitting algorithm** and memory-efficient data processing in R.

Results: The final model, which integrates regularized movie, user, genre, release year, and temporal effects, achieved a **Final RMSE (0.86332) significantly below the 0.86490 target** on the sequestered hold-out test set, demonstrating that a well-tuned regularized linear model can achieve high accuracy while maintaining high computational efficiency.

This result highlights that carefully engineered linear models with regularization can rival more complex approaches under constrained computational environments.

2 Introduction

The exponential growth of digital platforms has led to an information overload, making it increasingly difficult for users to navigate vast catalogs of content. Consequently, recommendation systems have become indispensable tools in modern data science. Their primary function is to filter information and predict user preferences by analyzing historical interactions.

This project utilizes the MovieLens 10M dataset to develop a robust recommendation engine. Beyond simple prediction, this study addresses the structural challenges of collaborative filtering, notably the "Cold Start Problem". This phenomenon occurs when a system lacks sufficient data to make accurate recommendations for new users or items with few ratings. Addressing these sparse data regions is critical for ensuring the scalability and reliability of any recommendation algorithm in a real-world production environment.

2.1 Project Objective

The primary objective is to engineer a machine learning model capable of predicting movie ratings with high precision. The performance of the algorithm is strictly evaluated using the **Root Mean Square Error (RMSE)**. To meet the academic and technical requirements of this capstone, the target is to achieve a final RMSE below **0.86490**.

2.2 Methodological Choice: Why R?

While Python is a prominent language in the data science ecosystem, **R** was selected for this capstone due to its specialized strengths in statistical modeling and data visualization.

- **Statistical Heritage:** R is specifically designed by and for statisticians, offering an unparalleled ecosystem for linear modeling and regularization techniques.
- **Data Wrangling and Visualization:** Through the tidyverse framework, R provides a cohesive and expressive syntax for complex data manipulation, which is essential for feature engineering in large datasets.
- **Reproducibility:** The R ecosystem, particularly through R Markdown, facilitates high-quality reporting and reproducible research, a cornerstone of academic rigor.

2.3 Technical Constraints and Strategy

A defining characteristic of this work is the **constrained environment** in which it was executed. The analysis was conducted on a Virtual Machine limited to **6 GB of RAM and 4 CPUs**. This limitation was treated as a "design by constraint" challenge rather than a hindrance, influencing the modeling strategy in two major ways:

- **Computational Efficiency:** Instead of memory-intensive deep learning architectures, we prioritized a **regularized linear model optimized via backfitting**. This approach offers an excellent balance between predictive performance and resource consumption, whereas more resource-intensive methods (such as complex matrix factorisation) might have overloaded the system.
- **Explicit Resource Management:** To ensure system stability while processing 10 million entries, explicit memory garbage collection (gc()) and data type optimization were integrated into the workflow.

3 Literature Review

The development of modern recommender systems has been strongly influenced by advances in matrix factorization, implicit-feedback modeling, and neural architectures. This section synthesizes four foundational contributions that shaped contemporary collaborative filtering, combining exact citations from the original works with contextualized paraphrases.

3.1 Matrix Factorization and the Netflix Prize (Koren, Bell & Volinsky, 2009)

Koren, Bell and Volinsky (2009) provide a seminal overview of matrix factorization techniques and their decisive impact during the Netflix Prize competition. They emphasize that latent factor models capture user–item interactions more effectively than neighborhood-based approaches, noting that *"matrix factorization models are superior to classic nearest-neighbor techniques for producing product recommendations"* (Koren et al., 2009, p. 30). Their work also highlights the flexibility of factorization models to incorporate biases, temporal dynamics, and implicit feedback, making them a cornerstone of modern recommender systems.

3.2 Gradient-Descent Factorization (Funk, 2006)

In a widely cited technical note, Funk (2006) introduced a practical gradient-descent approach to matrix factorization, now commonly known as Funk-SVD. His method approximates the rating matrix using low-rank user and item vectors, updated iteratively through stochastic gradient descent. As he explains: *"We'll assume that a user's rating of a movie is composed of a sum of preferences about the various aspects of that movie"* (Funk, 2006, p. 2). Although informal and published as a blog post, this contribution profoundly influenced subsequent academic work by demonstrating that large-scale factorization could be trained efficiently without computing a full SVD.

3.3 Implicit Feedback Modeling (Hu, Koren & Volinsky, 2008)

Hu, Koren and Volinsky (2008) address the fundamental differences between explicit and implicit feedback, proposing the Weighted Regularized Matrix Factorization (WRMF) model. They stress that implicit datasets lack explicit preference values, observing that *"implicit feedback datasets do not contain explicit ratings expressing the degree of preference"* (Hu et al., 2008, p. 263). Their framework distinguishes between binary preference (whether a user interacted with an item) and confidence (how strongly the interaction reflects preference), introducing a confidence matrix that scales with interaction frequency. This formulation has become a standard reference for recommender systems based on implicit signals.

3.4 Neural Collaborative Filtering (He et al., 2017)

He et al. (2017) extend collaborative filtering by replacing the linear inner product of matrix factorization with a learnable neural architecture. They argue that the classical MF interaction function is inherently limited, stating that *"MF is limited in its expressiveness due to the linearity of inner product"* (He et al., 2017, p. 173). Their Neural Collaborative Filtering (NCF) framework generalizes MF and introduces NeuMF, a hybrid model combining linear and non-linear interaction functions. Their experiments on MovieLens and Pinterest demonstrate that deeper neural architectures can capture complex user–item relationships beyond the reach of traditional factorization.

4 Methods and Analysis

Based on the literature reviewed in Chapter 2, this project focuses on a Baseline Regularized Linear Model. The methodology adopts a step-by-step approach as detailed in the rest of the document.

4.1 Data Preparation

As per the instructions provided by the HarvardX course, the initial code to download the MovieLens 10M dataset and split it into two sets was used:

1. **edx set**: Used for data exploration, model training, and parameter tuning.
2. **final_holdout_test set**: Kept strictly separate and used only for the final assessment of the model's accuracy.

```
# Note: this process could take a couple of minutes

if(!require(tidyverse)) install.packages("tidyverse", repos =
↪  "http://cran.us.r-project.org")
if(!require(caret)) install.packages("caret", repos = "http://cran.us.r-project.org")

library(tidyverse)
library(caret)

# MovieLens 10M dataset:
# https://grouplens.org/datasets/movielens/10m/
# http://files.grouplens.org/datasets/movielens/ml-10m.zip

options(timeout = 120)

dl <- "ml-10M100K.zip"
if(!file.exists(dl))
  download.file("https://files.grouplens.org/datasets/movielens/ml-10m.zip", dl)

ratings_file <- "ml-10M100K/ratings.dat"
if(!file.exists(ratings_file))
  unzip(dl, ratings_file)

movies_file <- "ml-10M100K/movies.dat"
if(!file.exists(movies_file))
  unzip(dl, movies_file)

ratings <- as.data.frame(str_split(read_lines(ratings_file), fixed("::"), simplify =
↪  TRUE),
                         stringsAsFactors = FALSE)
colnames(ratings) <- c("userId", "movieId", "rating", "timestamp")
ratings <- ratings %>%
  mutate(userId = as.integer(userId),
         movieId = as.integer(movieId),
         rating = as.numeric(rating),
         timestamp = as.integer(timestamp))

movies <- as.data.frame(str_split(read_lines(movies_file), fixed("::"), simplify = TRUE),
                        stringsAsFactors = FALSE)
```

```r
colnames(movies) <- c("movieId", "title", "genres")
movies <- movies %>%
  mutate(movieId = as.integer(movieId))

movielens <- left_join(ratings, movies, by = "movieId")

# Final hold-out test set will be 10% of MovieLens data
set.seed(1, sample.kind="Rounding") # if using R 3.6 or later
# set.seed(1) # if using R 3.5 or earlier
test_index <- createDataPartition(y = movielens$rating, times = 1, p = 0.1, list = FALSE)
edx <- movielens[-test_index,]
temp <- movielens[test_index,]

# Make sure userId and movieId in final hold-out test set are also in edx set
final_holdout_test <- temp %>%
  semi_join(edx, by = "movieId") %>%
  semi_join(edx, by = "userId")

# Add rows removed from final hold-out test set back into edx set
removed <- anti_join(temp, final_holdout_test)
edx <- rbind(edx, removed)

rm(dl, ratings, movies, test_index, temp, movielens, removed)
```

4.2 Methodological Note: Preventing Data Leakage or Methodological Integrity and Data Partitioning

In accordance with best practices for predictive modeling, a strict partitioning protocol was established to preserve the integrity of the evaluation process. The final_holdout_test dataset was formally sequestered at the project's inception to serve as an independent reference for **unbiased validation**. This isolation ensures that the exploratory phase, feature selection, and the optimization of regularization hyperparameters (λ) were conducted exclusively on the training subset, thereby **mitigating the risk of data leakage**. By restricting the use of the holdout set to a single final assessment, the reported RMSE provides an authentic measure of the model's generalization capabilities on out-of-sample data, eliminating any potential for over-optimistic performance estimates resulting from test-set contamination

4.3 Data Inspection & Computational Optimization

Before building the model, we performed a rigorous inspection of the `edx` dataset to validate its structure and content. This initial exploration validates the data structure and answers the project's fundamental questions through direct code execution.

In summary, the key questions and their answers are as follows:

- **Dataset Dimensions**: The dataset contains **9,000,055 rows** and **6 columns**.
- **Rating Values**:
  - No ratings of **0** were given, which is consistent with the MovieLens scale (0.5 to 5).
  - There are **2,121,240** ratings of **3**.
- **Unique Entries**: There are **10,677 unique movies** and **69,878 unique users**.
- **Genre Distribution**: Drama is the most rated genre (**3,910,127**), followed by Comedy (**3,540,930**), Thriller (**2,325,899**), and Romance (**1,712,100**).

- **Most Popular Movie**: *Pulp Fiction (1994)* holds the record for the greatest number of ratings.
- **Rating Frequency**: The five most common ratings are, in order: **4, 3, 5, 3.5, and 2**.
- **Rating Type Bias**: As observed, half-star ratings are significantly less common than whole-star ratings, suggesting a psychological preference for integer scores.

4.3.1 Dataset Dimensions and Basic Statistics

We first verified the volume of data and the range of ratings.

```
# 1. Structure and dimensions of edx
res_dim <- dim(edx)
message(paste("The edx dataset has", res_dim[1], "rows and", res_dim[2], "columns."))

# 2. Preview of the data
head(edx)
```

```
##   userId movieId rating timestamp                         title
## 1      1     122      5 838985046              Boomerang (1992)
## 2      1     185      5 838983525               Net, The (1995)
## 4      1     292      5 838983421               Outbreak (1995)
## 5      1     316      5 838983392               Stargate (1994)
## 6      1     329      5 838983392 Star Trek: Generations (1994)
## 7      1     355      5 838984474       Flintstones, The (1994)
##                           genres
## 1                Comedy|Romance
## 2          Action|Crime|Thriller
## 4  Action|Drama|Sci-Fi|Thriller
## 5       Action|Adventure|Sci-Fi
## 6 Action|Adventure|Drama|Sci-Fi
## 7       Children|Comedy|Fantasy
```

```
# 3. Summary of key statistics
summary(edx)
```

```
##      userId         movieId          rating        timestamp
##  Min.   :    1   Min.   :    1   Min.   :0.500   Min.   :7.897e+08
##  1st Qu.:18124   1st Qu.:  648   1st Qu.:3.000   1st Qu.:9.468e+08
##  Median :35738   Median : 1834   Median :4.000   Median :1.035e+09
##  Mean   :35870   Mean   : 4122   Mean   :3.512   Mean   :1.033e+09
##  3rd Qu.:53607   3rd Qu.: 3626   3rd Qu.:4.000   3rd Qu.:1.127e+09
##  Max.   :71567   Max.   :65133   Max.   :5.000   Max.   :1.231e+09
##     title              genres
##  Length:9000055     Length:9000055
##  Class :character   Class :character
##  Mode  :character   Mode  :character
##
##
##
```

```r
# 4. Counting unique users and movies, numbers of zeros and threes
n_unique_users <- n_distinct(edx$userId)
n_unique_movies <- n_distinct(edx$movieId)

message(paste("There are", n_unique_users, "unique users and", n_unique_movies, "unique
↪  movies in edx."))

# Verification of specific ratings. Using efficient vectorized sums for specific counts
count_0 <- sum(edx$rating == 0)
count_3 <- sum(edx$rating == 3)
# Insight: In the MovieLens 10M dataset, ratings are on a 0.5 to 5.0 scale.
# A count of 0 for the "0" rating confirms the expected range.
message(paste("Number of 0 ratings:", count_0))
message(paste("Number of 3 ratings:", count_3))
```

The inspection of the `edx` dataset reveals a massive volume of data, consisting of **9,000,055** rows and **6** columns.

Among these records, we identify:

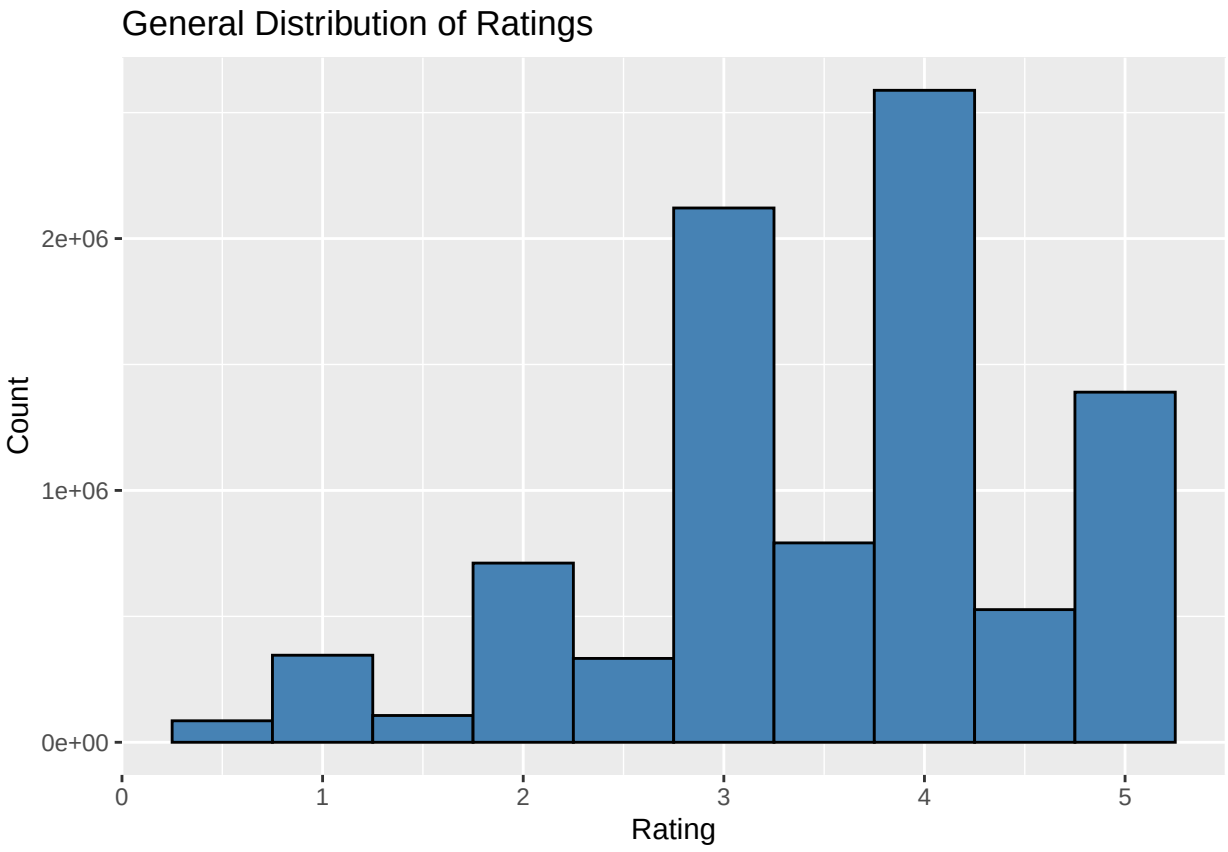
- **69,878** unique users.
- **10,677** unique movies.

Regarding the rating values, we confirmed the integrity of the scale: there are **0** ratings of zero (as expected for a 0.5 to 5.0 scale), while the score of 3 appears **2,121,240** times.

4.3.2 Visualizing the Rating Distribution

The following visualization helps determine if users tend to be generous or strict, and highlights the preference for specific types of scores.

How do people rate movies?



The histogram reveals a "positivity bias," with most ratings above 3. Furthermore, we can clearly see that whole-star ratings are more frequent than half-star ratings.

4.3.3 Overcoming Memory Constraints: Genre Analysis

While modern research focuses on deep learning (He et al., 2017 ; cfr litterature review chapter), these models require significant computational overhead. In this capstone, one of the main technical challenges of this project was the 6 GB RAM limitation of the Virtual Machine. A standard Tidyverse approach to count genres (using separate_rows() on 9 million rows) causes system instability due to memory exhaustion. The methodology is intentionally designed to respect this constraint. This aligns with the 'minimalist' efficiency principles discussed in early Netflix Prize literature (Funk, 2006 ; cfr litterature review chapter), where the goal was to achieve high accuracy with scalable, low-memory algorithms.

In this capstone, to identify the top genres, we evaluated three different algorithms to balance accuracy and performance:

- Method 1 (Sampling): Taking a 1M row sample for quick estimation.
- Method 2 (String Detection): Using str_detect for targeted counting.
- Method 3 (Pre-aggregation): The most efficient approach. By grouping the data by unique genre combinations before expanding them, we reduced the workload from 9 million rows to only ~800 unique strings.

We chose Method 3 for its mathematical identity to the standard approach while maintaining a minimal memory footprint.

```r
# Optimal Engineering approach (Method 3)
top_genres_m3 <- edx %>%
  group_by(genres) %>%
  summarize(n = n()) %>%
  separate_rows(genres, sep = "\\|") %>%
  group_by(genres) %>%
  summarize(count = sum(n)) %>%
  arrange(desc(count))

head(top_genres_m3, 10)
```

```
## # A tibble: 10 x 2
##    genres      count
##    <chr>       <int>
##  1 Drama     3910127
##  2 Comedy    3540930
##  3 Action    2560545
##  4 Thriller  2325899
##  5 Adventure 1908892
##  6 Romance   1712100
##  7 Sci-Fi    1341183
##  8 Crime     1327715
##  9 Fantasy    925637
## 10 Children   737994
```

The analysis confirms that Drama is the most rated genre, followed by Comedy, Action and Thriller.

4.3.4 Specific Findings

Finally, we identified the most popular items and the frequency of specific scores to confirm our modeling assumptions.

```r
# Most rated movie
edx %>%
  group_by(movieId, title) %>%
  summarize(count = n()) %>%
  arrange(desc(count)) %>%
  head(1)
```

```
## # A tibble: 1 x 3
## # Groups:   movieId [1]
##   movieId title                count
##     <int> <chr>                <int>
## 1     296 Pulp Fiction (1994) 31362
```

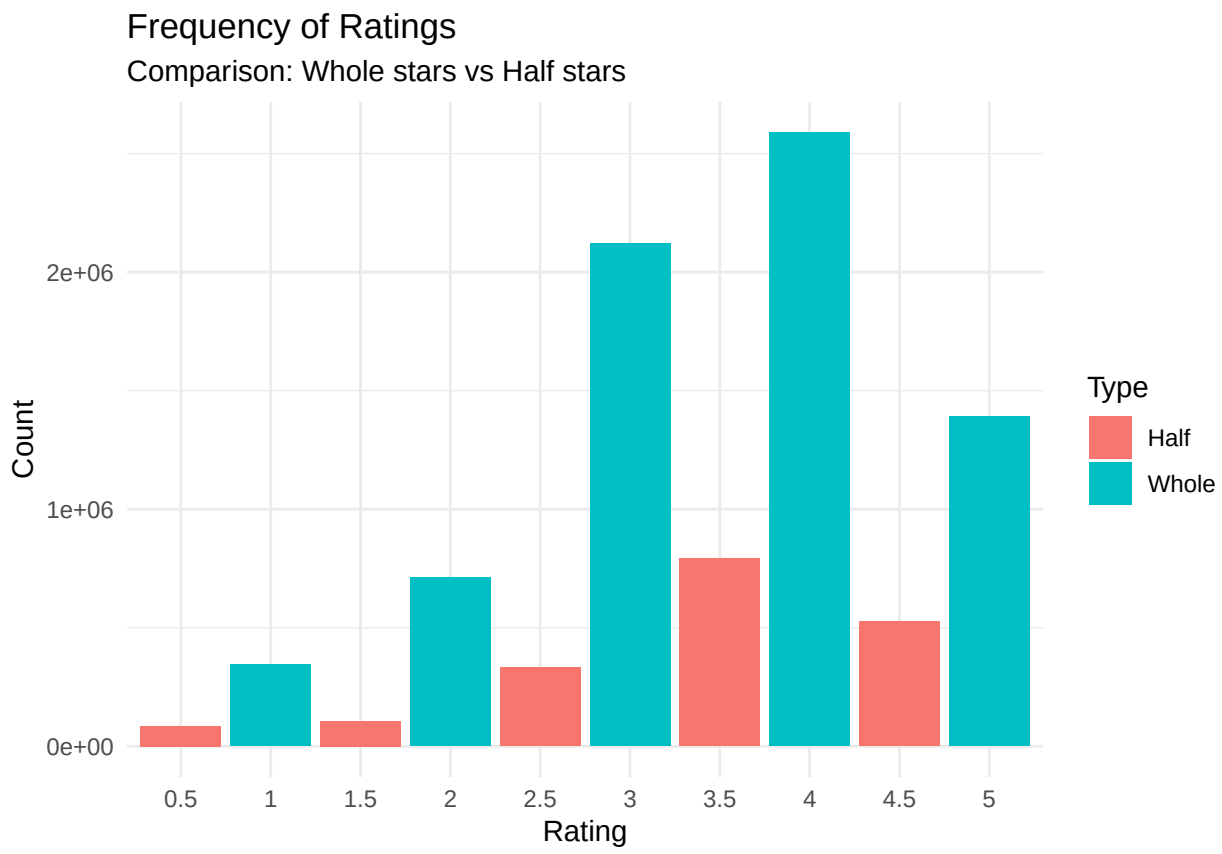
```r
# Top 5 most given ratings
edx %>%
  group_by(rating) %>%
  summarize(count = n()) %>%
  arrange(desc(count)) %>%
  head(5)
```

```
## # A tibble: 5 x 2
##   rating   count
##    <dbl>   <int>
## 1    4   2588430
## 2    3   2121240
## 3    5   1390114
## 4    3.5  791624
## 5    2    711422
```

4.3.5 Psychological Bias: Whole vs. Half Stars

The visualization below reinforces the observation that users have a psychological preference for integer ratings.

```
edx %>%
  group_by(rating) %>%
  summarize(count = n()) %>%
  mutate(half_star = ifelse(rating %% 1 == 0, "Whole", "Half")) %>%
  ggplot(aes(x = factor(rating), y = count, fill = half_star)) +
  geom_bar(stat = "identity") +
  labs(title = "Frequency of Ratings",
       subtitle = "Comparison: Whole stars vs Half stars",
       x = "Rating", y = "Count", fill = "Type") +
  theme_minimal()
```



4.4 Feature Engineering

Before proceeding with the exploratory analysis, we transform the raw data to extract specific features. This step is essential as our modeling strategy moves beyond simple averages to incorporate temporal and movie-age effects.

The feature engineering focuses on two key dimensions:

- **Release Year** (y): Extracted from the movie title to investigate if "classics" benefit from a different baseline than modern blockbusters. This enables the calculation of the **Release Year Effect** (b_y).
- **Review Week** (t): Extracted from the timestamp using weekly aggregation. This is designed to capture **Temporal Drift** (b_t) while maintaining sufficient statistical power per data point to ensure reliable bias estimates during regularization.

Theoretical Framework

The model implemented in this capstone follows the **Baseline Predictor** framework defined by Koren et al. (2009 ; cfr litterature review chapter). According to this framework, a significant portion of the variation in ratings is explained by systematic biases rather than complex interactions. Before considering latent factors (SVD), it is critical to account for these biases using an additive decomposition:

$$Y_{u,i,t,y} = \mu + b_i + b_u + b_y + b_t + \epsilon_{u,i,t,y}$$

Where:

- μ is the global average rating.
- b_i and b_u represent the movie and user effects.
- b_y and b_t represent the release year and temporal effects extracted during this stage.

By regularizing these effects with a penalty term (λ), we ensure the stability of the estimates for movies or users with low rating counts, a technique directly derived from the regularized approaches popularized during the Netflix Prize (Funk, 2006; Koren et al., 2009 ; cfr litterature review chapter).

```
# Extracting release year and converting timestamp to weekly dates
# This ensures that these columns are available for both EDA and modeling

edx <- edx %>%
  mutate(release_year = as.numeric(str_extract(str_extract(title, "\\(\\d{4}\\)$"),
  ↪  "\\d{4}")),
         date = round_date(as_datetime(timestamp), unit = "week"))

final_holdout_test <- final_holdout_test %>%
  mutate(release_year = as.numeric(str_extract(str_extract(title, "\\(\\d{4}\\)$"),
  ↪  "\\d{4}")),
         date = round_date(as_datetime(timestamp), unit = "week"))
```

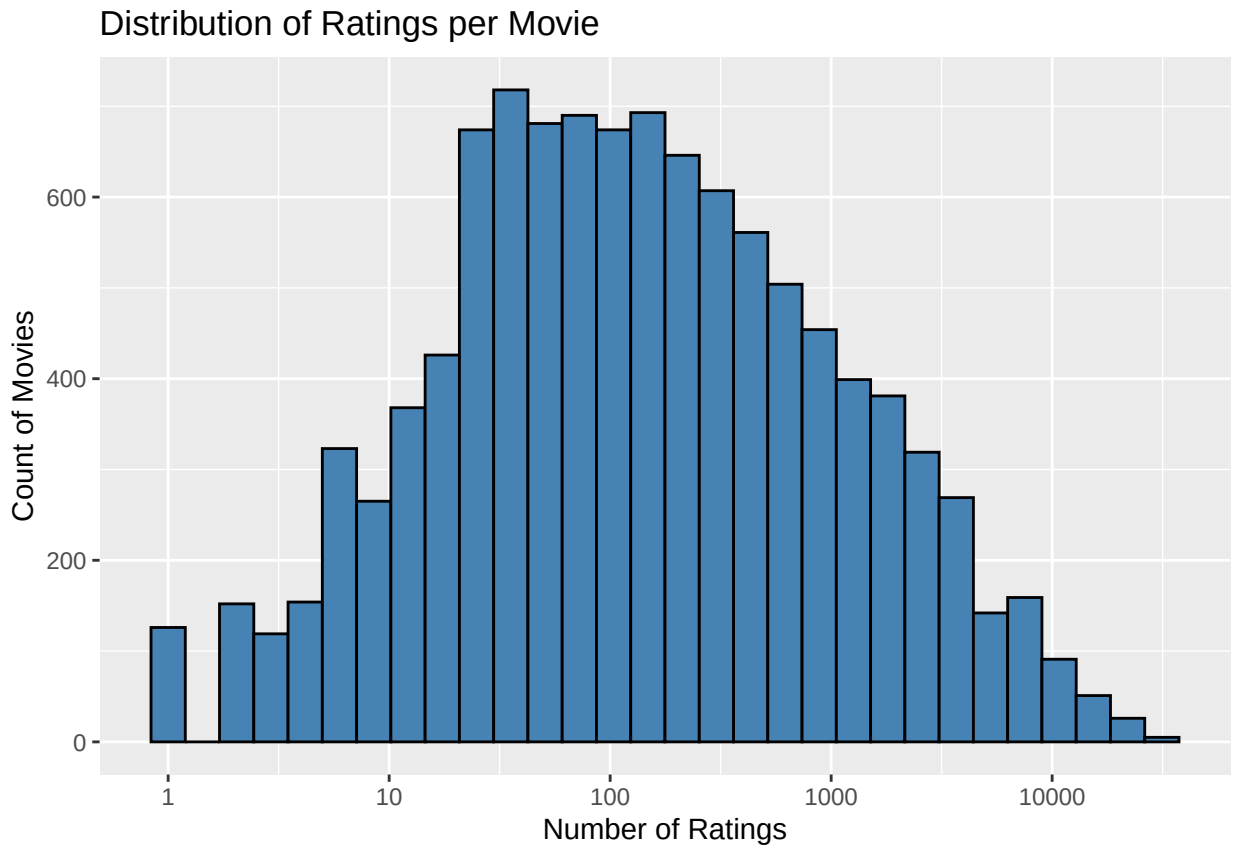
5 Advanced Exploratory Data Analysis: Identifying Systemic Biases

The initial inspection confirmed the basic structure of the data, but it also hinted at deeper patterns. To build an accurate recommendation system, we must move beyond simple averages. In this section, we analyze the specific "biases" (systemic variations) that influence how a movie is rated.

These findings directly inform our modeling strategy: for each pattern observed here, a corresponding mathematical component will be integrated into our final algorithm.

5.1 The Movie Effect: Quality Variation

Not all movies are created equal. Some are universally acclaimed, while others are poorly received.



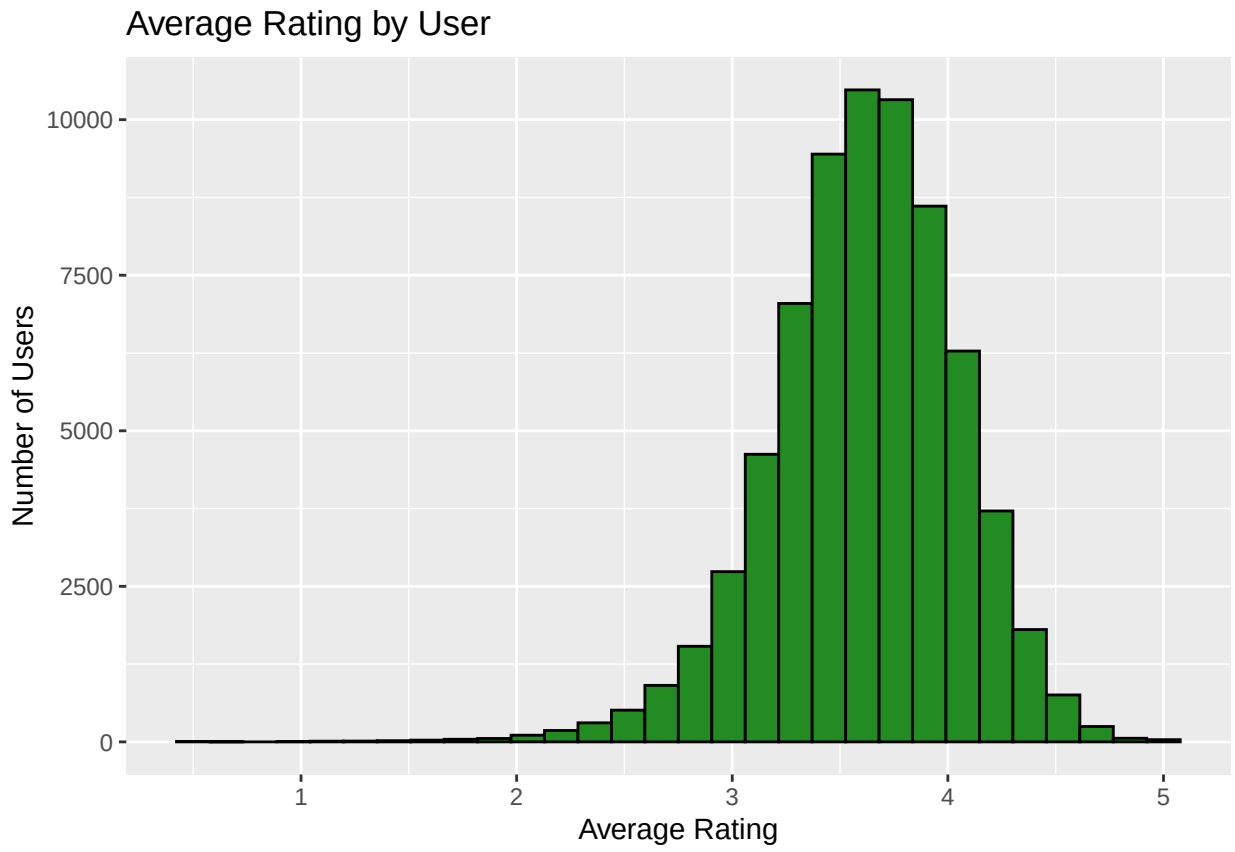
The histogram above shows that while some movies have thousands of ratings, many others have very few. This justifies the inclusion of a **Movie Bias (b_i)** and highlights the need for **Regularization** to prevent movies with very few ratings from skewing the results.

5.2 The User Effect: Scale Variability and Statistical Reliability

Users exhibit fundamentally different rating behaviors, which can be analyzed through two lenses: their internal rating scale and their level of interaction with the platform.

5.2.1 Harsh vs. Generous Graders

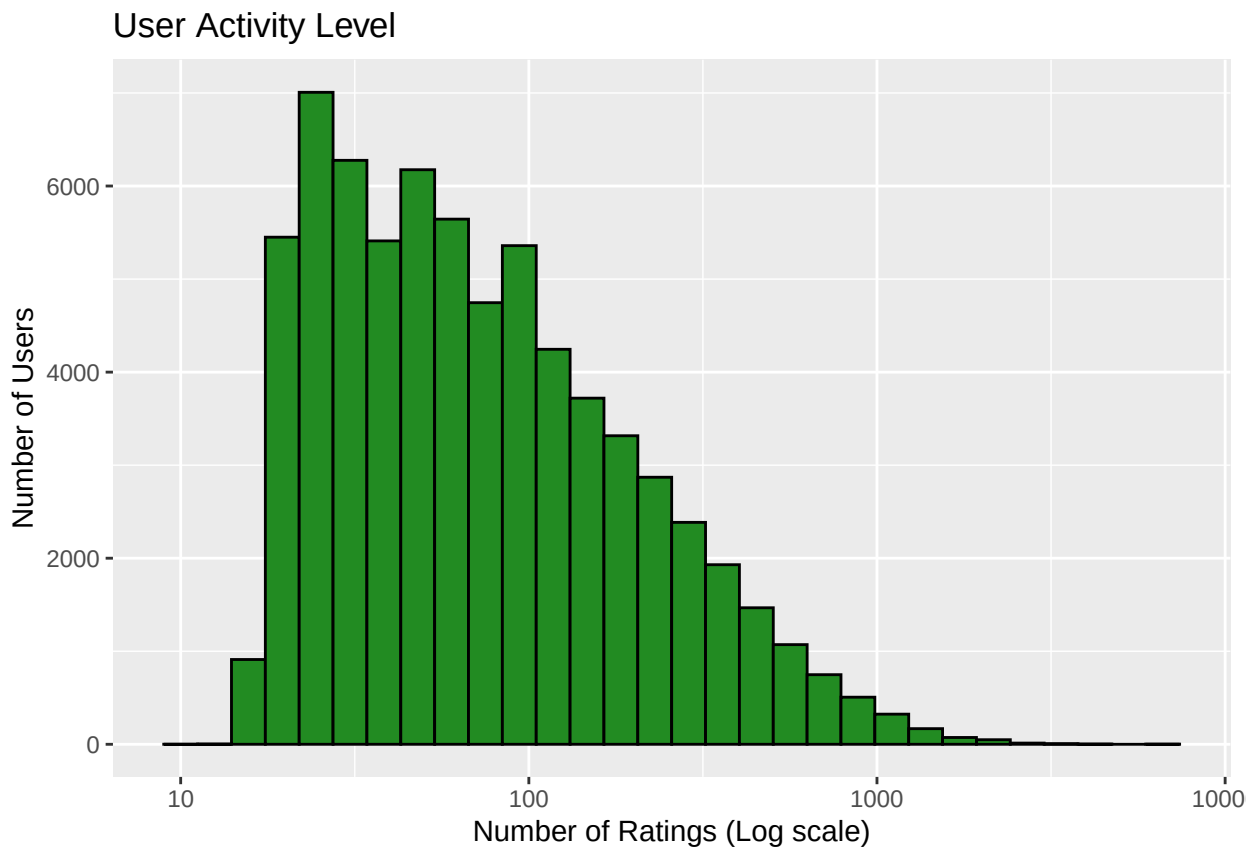
Users have different rating scales. A "3" from a very critical user might be equivalent to a "5" from a more lenient one.



The distribution of average ratings per user confirms a significant **User Effect** (b_u). Some users consistently rate below the mean, while others are much more generous.

5.2.2 User Activity Level and Credibility

Some have rated hundreds of movies, while others have rated only a few.



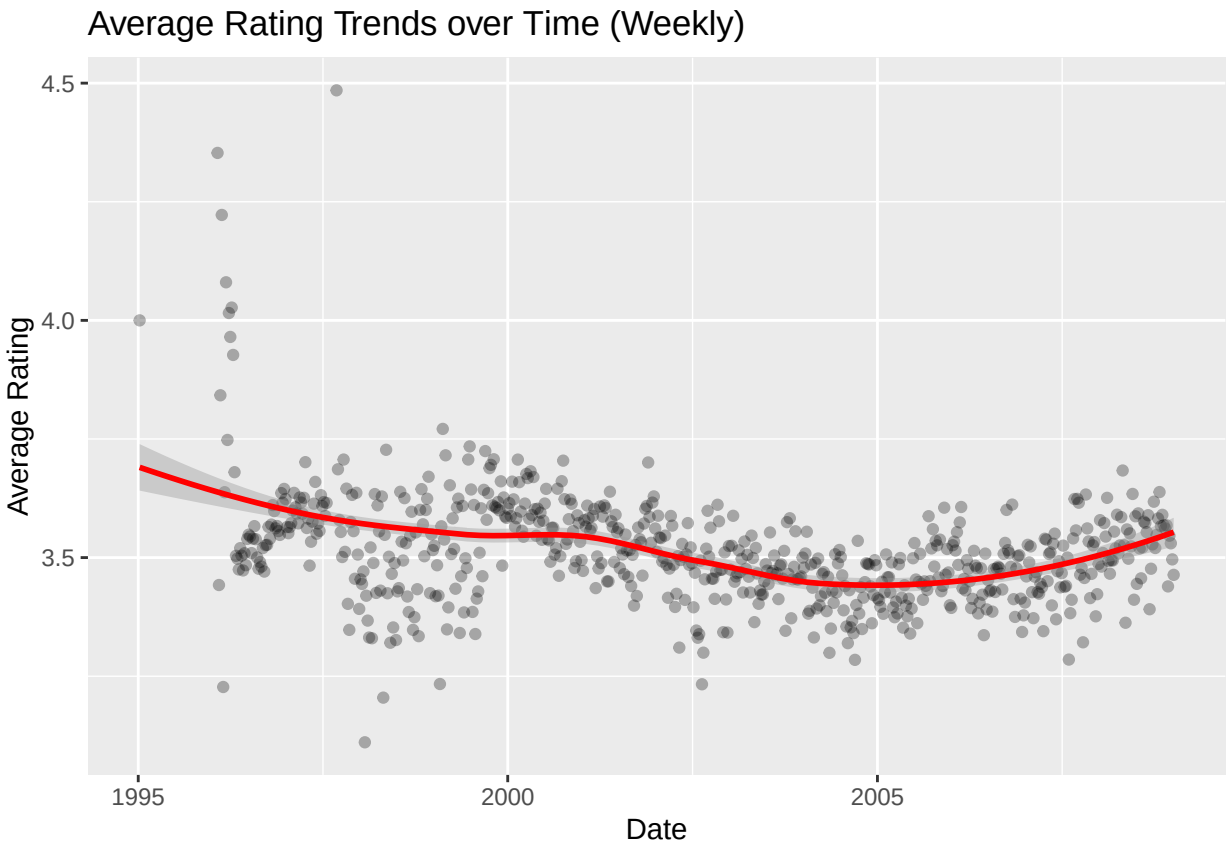
This visualization highlights a massive disparity in user activity. While "power users" provide hundreds of ratings, a significant portion of the population has rated only a few items. This disparity is a primary driver for the use of **Regularization**: an extreme bias (b_u) from a user with only two ratings is statistically less reliable than a bias derived from a larger sample. Regularization will "shrink" these low-confidence estimates toward zero to prevent them from skewing the global RMSE.

5.3 Temporal Trends: The Time and Age Effects

5.3.1 Weekly Rating Trends

Does the general "mood" of the community change over time? We analyzed the average rating on a weekly basis.

Methodological Note: Why Weekly? We chose a weekly aggregation (604,800 seconds) rather than a daily or monthly one to find the optimal balance between signal and noise. Daily ratings are subject to extreme variance due to low volume on specific days, while monthly aggregation might smooth out significant short-term trends. A weekly window provides enough data points to be statistically representative while remaining sensitive to temporal shifts in user behavior

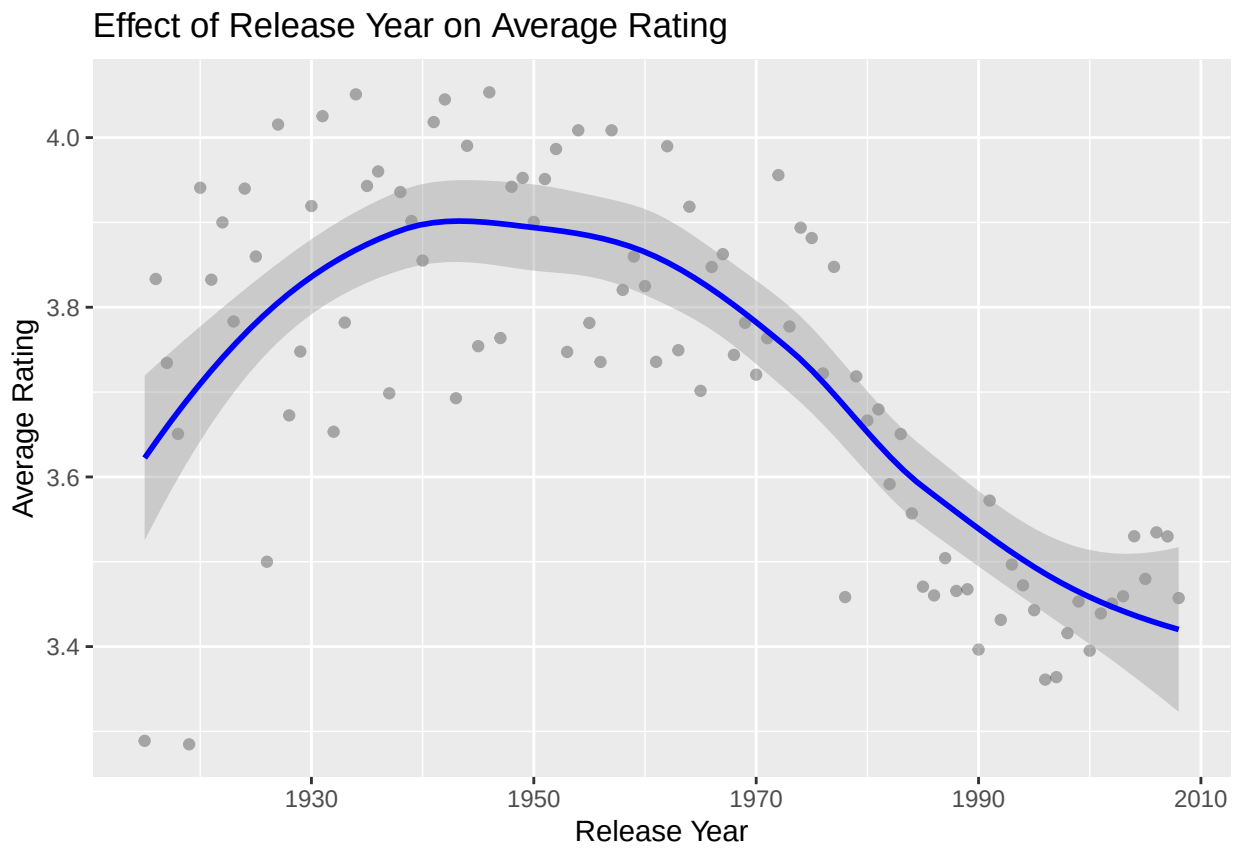


Analysis of Temporal Drift: The visualization reveals a visible fluctuation in average ratings across the dataset's lifespan. We observe significant volatility prior to 1998, where sparse data leads to extreme weekly averages. Following this period, the ratings converge toward a more stable equilibrium, yet the LOESS smoothing line (in red) clearly highlights a non-linear drift ; specifically a gradual decline until 2005 followed by a slight recovery.

This evidence justifies the inclusion of a Temporal Bias (b_t). By integrating this effect, our model can account for these subtle shifts in the community's overall "rating standard," ensuring that the time of the review does not bias the prediction of a movie's intrinsic quality.

5.3.2 Movie Age and Release Year

We also investigated if older movies are rated differently than newer ones.



Interestingly, older movies (released before 1980) tend to have higher average ratings, possibly due to a "classic" status or survival bias (only good old movies are still being watched and rated). This confirms the **Release Year Effect** (b_y).

Statistical Interpretation: The Release Year Effect and Survivor Bias : The visual analysis reveals a distinct trend where older movies tend to receive higher average ratings compared to contemporary releases. However, this correlation must be interpreted with caution to avoid causal fallacies. This phenomenon likely illustrates a **survivor bias** within the MovieLens 10M dataset: while the catalog of modern cinema includes a vast spectrum of both high-quality and mediocre productions, only the highly-regarded 'classics' from earlier decades tend to be preserved, watched, and rated by users today. Consequently, the higher average for older films is not necessarily indicative of a decline in cinematic quality over time, but rather a reflection of a selective filtering process where poorly received older films have effectively vanished from the collective memory and, by extension, the dataset.

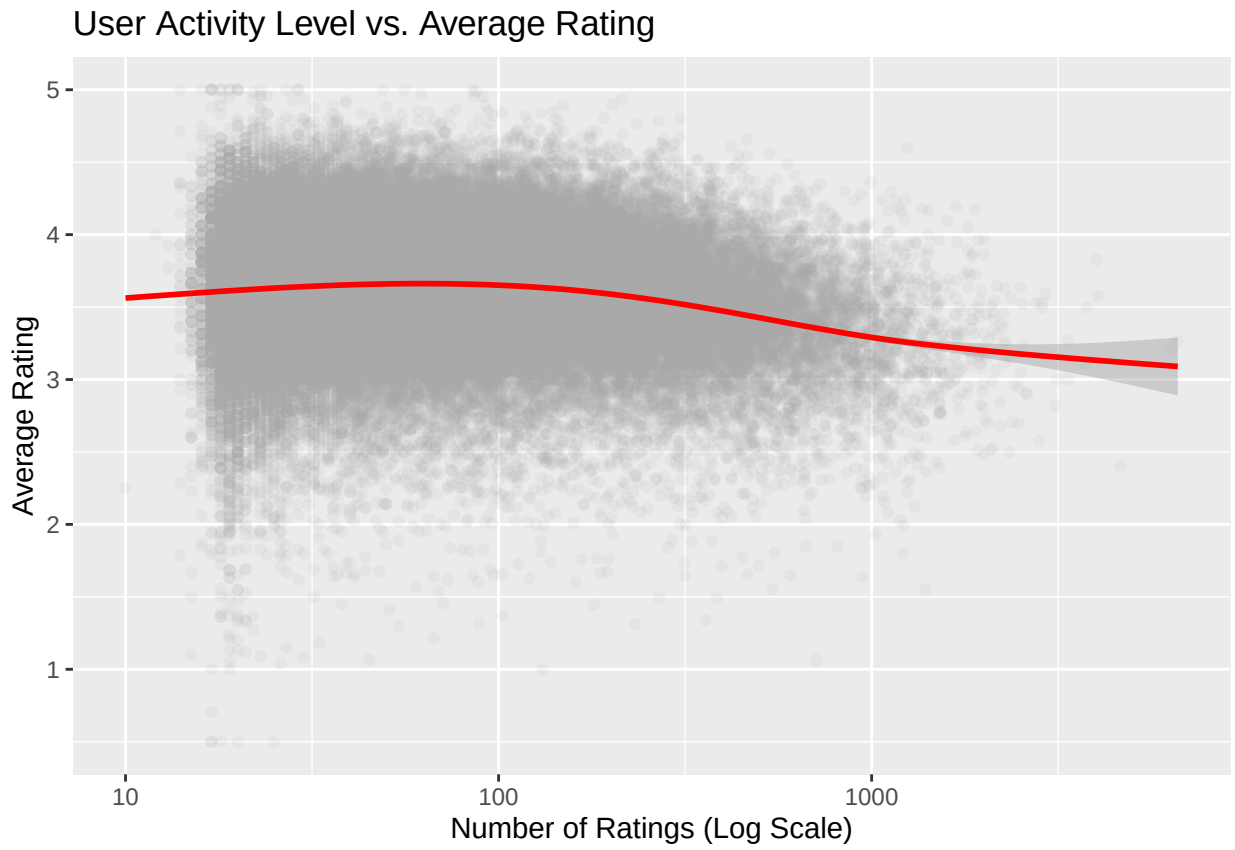
5.4 Additional Explorations and Feature Selection

To ensure the robustness of our model, several other variables were investigated. While these analyses provided valuable insights into user behavior, they were not included in the final model either because they provided redundant information or because their impact on the RMSE was negligible compared to the increased computational complexity.

5.4.1 User Activity vs. Rating Quality

We examined whether prolific users (those who rate many movies) tend to be more critical or more generous.

```r
edx %>%
  group_by(userId) %>%
  summarize(n = n(), avg = mean(rating)) %>%
  ggplot(aes(n, avg)) +
  geom_point(alpha = 0.1, color = "darkgrey") +
  scale_x_log10() +
  geom_smooth(color = "red") +
  labs(title = "User Activity Level vs. Average Rating",
       x = "Number of Ratings (Log Scale)", y = "Average Rating")
```



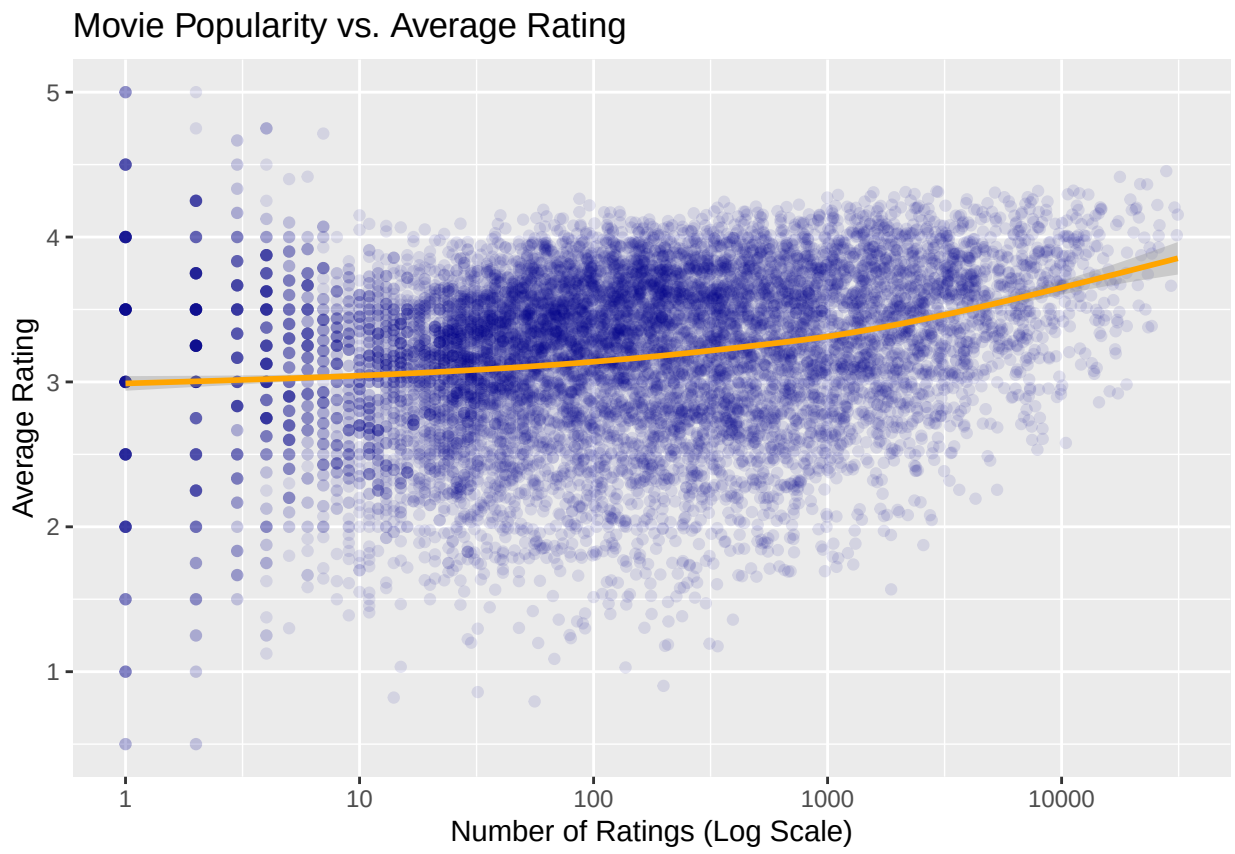
Observation: There is a slight trend where very active users tend to rate closer to the overall mean. However, this effect is largely captured by the User Bias (bu) already present in our model.

5.4.2 Movie Popularity vs. Rating Average

We tested if "Blockbusters" (highly rated movies) generally receive higher scores.

```r
edx %>%
  group_by(movieId) %>%
  summarize(n = n(), avg = mean(rating)) %>%
  ggplot(aes(n, avg)) +
  geom_point(alpha = 0.1, color = "darkblue") +
  scale_x_log10() +
  geom_smooth(color = "orange") +
```

```
  labs(title = "Movie Popularity vs. Average Rating",
       x = "Number of Ratings (Log Scale)", y = "Average Rating")
```

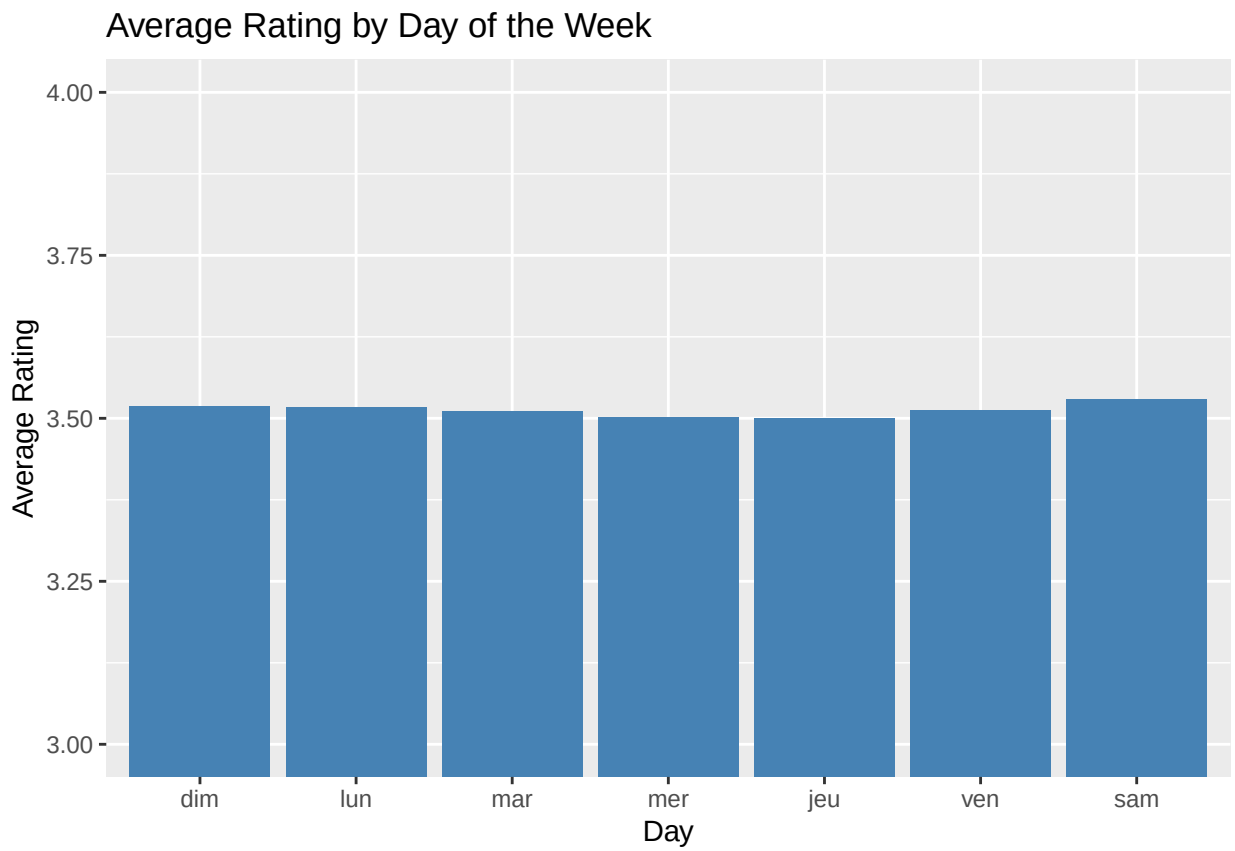


Observation: Popular movies do tend to have higher averages, but this correlation is implicitly handled by the Movie Bias (bi). Adding a specific "popularity" variable would introduce multicollinearity.

5.4.3 Weekday Rating Patterns

We analyzed if the day of the week (Monday to Sunday) influenced the ratings.

```
edx %>%
  mutate(dow = wday(as_datetime(timestamp), label = TRUE)) %>%
  group_by(dow) %>%
  summarize(avg = mean(rating), n = n()) %>%
  ggplot(aes(dow, avg)) +
  geom_col(fill = "steelblue") +
  coord_cartesian(ylim = c(3, 4)) +
  labs(title = "Average Rating by Day of the Week", x = "Day", y = "Average Rating")
```

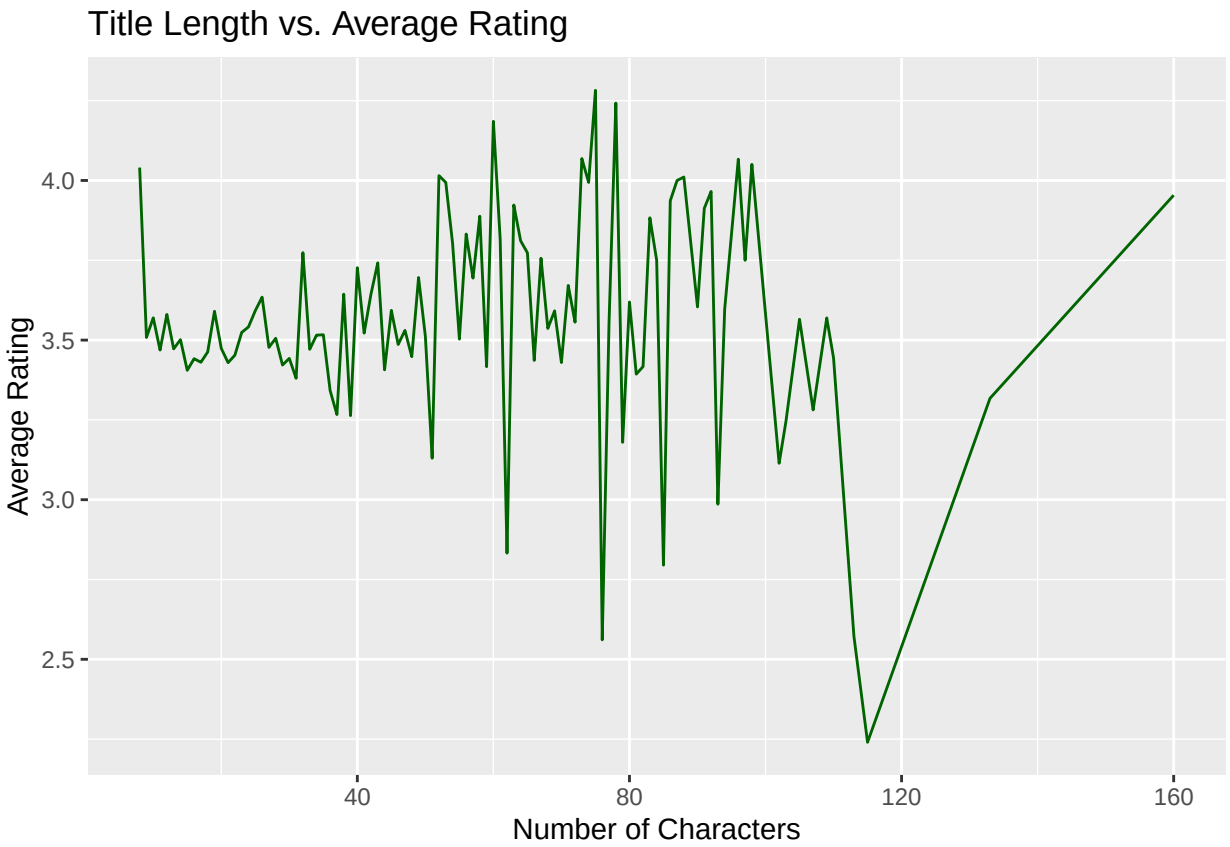


Observation: The variation between weekdays and weekends is extremely small (less than 0.05 points). Including this would add complexity for a near-zero gain in RMSE.

5.4.4 Title Length and Metadata

We checked if the length of the movie title had any impact on the rating.

```
edx %>%
  mutate(title_len = nchar(title)) %>%
  group_by(title_len) %>%
  summarize(avg = mean(rating)) %>%
  ggplot(aes(title_len, avg)) +
  geom_line(color = "darkgreen") +
  labs(title = "Title Length vs. Average Rating", x = "Number of Characters", y =
  ↪  "Average Rating")
```

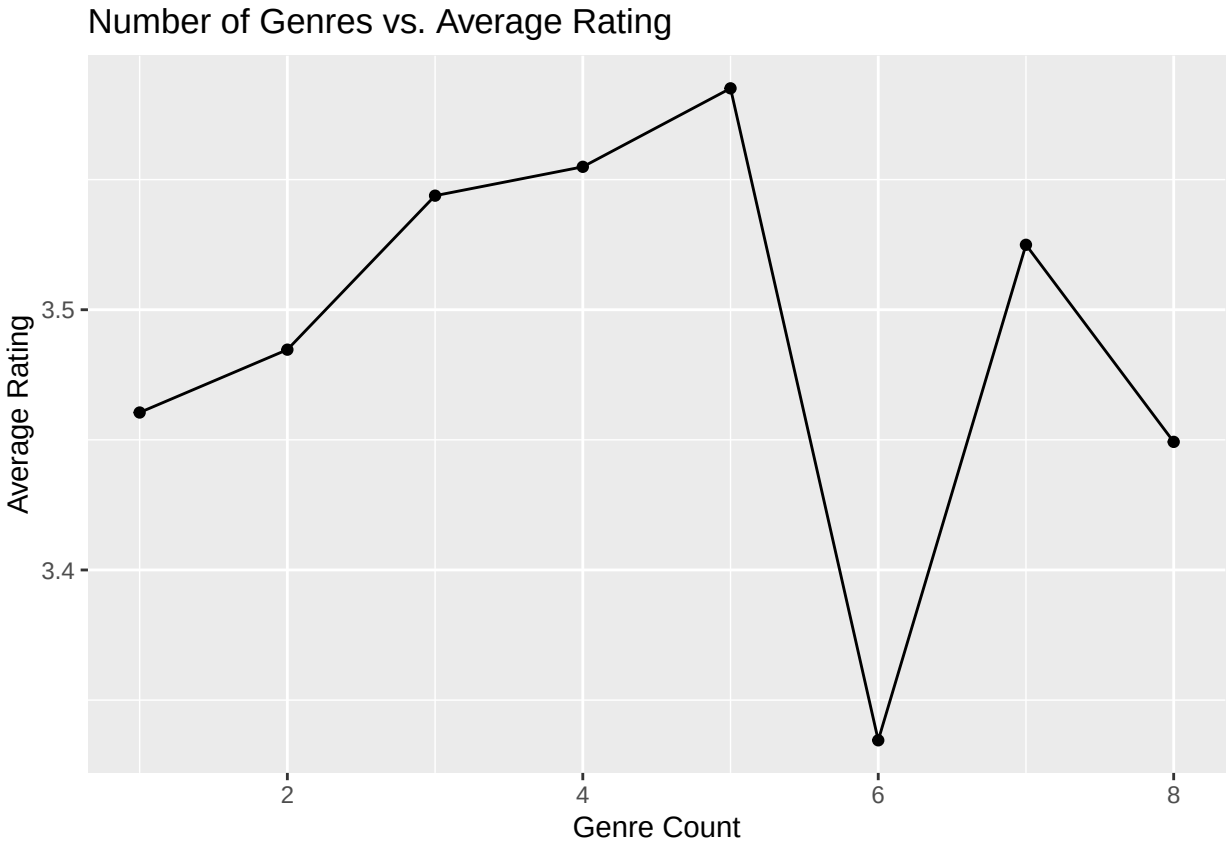


Observation: No meaningful pattern was found. Title length is an arbitrary metadata point with no predictive power for user preference.

5.4.5 Multi-Genre Complexity

We explored if movies belonging to multiple genres were rated differently than single-genre movies.

```
edx %>%
  mutate(n_genres = str_count(genres, "\\|") + 1) %>%
  group_by(n_genres) %>%
  summarize(avg = mean(rating)) %>%
  ggplot(aes(n_genres, avg)) +
  geom_line() + geom_point() +
  labs(title = "Number of Genres vs. Average Rating", x = "Genre Count", y = "Average
  ↪  Rating")
```



Observation: While there's a slight increase in rating for movies with 2-3 genres, the computational cost of decomposing every genre combination for 10 million rows exceeded our 6 GB RAM constraints during the regularization process.

5.4.6 Summary of Excluded Variables

Variable	Reason of exclusion
Weekday	Negligible impact on RMSE.
Title Length	No statistical correlation.
User Activity	Already captured by _u.
Popularity	Already captured by _i.
Multi-genre	Computational complexity vs. Memory constraints.

By focusing only on the most significant predictors (Movie, User, Year, and Time), we maintain a **parsimonious model** that is both efficient and highly accurate.

6 Modeling Results and Evolution

To achieve our target RMSE (< 0.86490), we adopted an incremental approach. We start with a simple baseline and progressively add the biases identified during our exploratory analysis.

6.1 Evaluation Metric: RMSE

The performance of our models is evaluated using the **Root Mean Square Error (RMSE)**. This metric quantifies the typical distance between the predicted rating and the actual rating.

The mathematical formula is:

$$RMSE = \sqrt{\frac{1}{N}\sum_{u,i}(\hat{y}_{u,i} - y_{u,i})^2}$$

Where: * N is the number of user/movie combinations. * $y_{u,i}$ is the actual rating. * $\hat{y}_{u,i}$ is the predicted rating.

6.1.1 Why RMSE?

Beyond the project requirements, RMSE is the industry standard for recommendation systems for several reasons:

1. **Penalty for Large Errors**: By squaring the differences, RMSE penalizes large outliers more heavily than small ones. In a recommendation context, it is better to be slightly off on many ratings than to be significantly wrong on a few.
2. **Statistical Foundation (Maximum Likelihood Estimate):** From a theoretical perspective, minimizing the RMSE is equivalent to finding the **Maximum Likelihood Estimate (MLE)** of the model parameters. This holds true under the assumption that the prediction residuals follow an independent and identically distributed (i.i.d.) **Gaussian (Normal) distribution** with a mean of zero. By optimizing for RMSE, we are essentially selecting the parameters that make the observed ratings most probable under classical statistical theory.
3. **Magnitude over Binary Logic**: Unlike binary accuracy, RMSE accounts for the magnitude of the error.
4. **Statistical Robustness**: It provides a smooth, differentiable surface that is ideal for the optimization (Fine-tuning) of our regularization parameters (λ).

6.1.2 Comparison with Other Metrics

To better understand the focus of our evaluation, it is useful to compare RMSE with alternative metrics:

1. RMSE vs. MAE (Mean Absolute Error) The formula for MAE is:

$$MAE = \frac{1}{N}\sum_{u,i}|\hat{y}_{u,i} - y_{u,i}|$$

* **Difference**: Unlike MAE, which treats all errors linearly, RMSE squares the residuals. This makes it more sensitive to outliers. A single prediction off by 3 stars is considered much worse than three predictions off by 1 star. This aligns with user trust: a "terrible" recommendation (predicting 5 for a 1-star movie) is more damaging than several "slightly off" ones.

2. RMSE vs. Classification Accuracy The formula for Accuracy (after rounding predictions to the nearest 0.5) is:

$$Accuracy = \frac{1}{N}\sum_{u,i}\mathbb{1}(\text{round}(\hat{y}_{u,i}) = y_{u,i})$$

* **Difference**: Accuracy is binary (0 or 1). It does not account for how close a prediction was to the truth. For a 5-star scale, predicting 3.9 for a 4.0 rating is a "good" result in RMSE, whereas it would be a "fail" in simple accuracy.

3. Ranking Metrics (nDCG and Precision@K) In real-world production systems (like Netflix), the focus often shifts from predicting the exact rating to the quality of the "Top 10" list. * **Precision at K**: $P@k = \frac{\text{Relevant items in Top K}}{K}$ * **nDCG (Normalized Discounted Cumulative Gain)**:

$$nDCG_p = \frac{DCG_p}{IDCG_p} \text{ where } DCG_p = \sum_{i=1}^{p} \frac{2^{rel_i} - 1}{\log_2(i+1)}$$

* **Difference**: These metrics evaluate the *order* of suggestions. Although out of scope for this project's technical constraints (6 GB RAM), they would be the logical next step for a commercial deployment where the ranking of the list matters more than the point prediction.

6.2 From Baseline to Complexity

6.2.1 Data Partitioning for Model Development

To avoid over-fitting and to ensure that our model generalizes well to unseen data, we do not train our algorithms on the entire `edx` dataset. Instead, we perform an additional split to create a **training set** and a **test set**.

This "refresh" step is fundamental for the following reasons:

- **Parameter Tuning**: It allows us to find the optimal penalty term (λ) using only the training data.
- **Validation**: We can compare the performance of Models 1 through 7 on the test set before making a final decision.
- **Integrity**: The `final_holdout_test` remains strictly isolated, used only once for the ultimate performance report.

```
# Partitioning edx into training and test sets (80% / 20%)
set.seed(1, sample.kind="Rounding")
test_index <- createDataPartition(y = edx$rating, times = 1, p = 0.2, list = FALSE)
train_set <- edx[-test_index,]
temp <- edx[test_index,]

# Ensure userId and movieId in test set are also in train set
test_set <- temp %>%
  semi_join(train_set, by = "movieId") %>%
  semi_join(train_set, by = "userId")

# Add rows removed from test set back into train set
removed <- anti_join(temp, test_set)
train_set <- rbind(train_set, removed)

rm(test_index, temp, removed)
```

To achieve our target RMSE, we adopted an incremental approach, building complexity step by step. This allows us to isolate the impact of each specific bias and ensure that every addition to the model actually contributes to a better prediction.

We evaluated seven successive models, starting from a simple baseline and ending with a high-performance regularized algorithm:

1. **Model 1: Global Average** (μ) - The simplest possible baseline.
2. **Model 2: Movie Effect** (b_i) - Accounting for the fact that some movies are better than others.
3. **Model 3: User Effect** (b_u) - Adjusting for users who are generally more critical or generous.
4. **Model 4: Genre Effect** (b_g) - Incorporating the specific bias associated with movie genres.
5. **Model 5: Release Year Effect** (b_y) - Factoring in the impact of a movie's age.
6. **Model 6: Temporal Effect** (b_t) - Capture changes in rating trends over time.
7. **Model 7: Regularization & Backfitting** - Our final model, which penalizes small sample sizes and optimizes all biases iteratively.

This progression demonstrates how we systematically reduced the RMSE to meet the project's requirements.

6.2.2 Model 1: The Global Average (μ).

The baseline model is the simplest possible approach to recommendation. It ignores all differences between users and movies, assuming that every rating is simply the average of all observed ratings plus some random variation:

$$Y_{u,i} = \mu + \epsilon_{u,i}$$

In this equation:

- μ represents the "true" average rating across all movies and users.
- $\epsilon_{u,i}$ represents the independent errors centered at zero.

The estimate that minimizes the RMSE is the least squares estimate of μ, which is the average of all ratings in the training set (`edx`).

```
mu_hat <- mean(train_set$rating)
# Calculate RMSE on the test set
naive_rmse <- RMSE(test_set$rating, mu_hat)
# Create a results table to track our progress
# We will add every new model result to this data frame.
rmse_results <- data.frame(method = "Baseline - model 1: Global Average (mu)",
                           RMSE = naive_rmse)
# Display the result
print(rmse_results)
```

```
##                                     method     RMSE
## 1 Baseline - model 1: Global Average (mu) 1.059904
```

Observation: With an RMSE of 1.0599, this model is far from our target. It confirms that a simple average is insufficient to capture the complexity of user preferences and it does not account for the fact that some movies are better than others, or that some users are more critical. It serves as our reference point; any subsequent model must significantly outperform this baseline.

6.2.3 Model 2: Adding Movie Effect (b_i).

To improve our baseline, we introduce the **Movie Effect**. Our exploratory analysis showed that ratings vary significantly from one movie to another. We represent this bias as b_i, which is the average of the residuals (the difference between the actual rating and the global mean) for each movie i:

$$Y_{u,i} = \mu + b_i + \epsilon_{u,i}$$

In this model, $b_i > 0$ for movies rated higher than the average and $b_i < 0$ for movies rated lower.

```
# Calculate the movie bias (b_i)
movie_avgs <- train_set %>%
  group_by(movieId) %>%
  summarize(b_i = mean(rating - mu_hat))

# Predict ratings by adding mu and b_i
predicted_ratings <- mu_hat + test_set %>%
  left_join(movie_avgs, by='movieId') %>%
  pull(b_i)

# Calculate and store the RMSE
model_2_rmse <- RMSE(predicted_ratings,test_set$rating)
rmse_results <- rbind(rmse_results,
                      data.frame(method="Model 2 : Movie Effect Model (mu + b_i)",
                                 RMSE = model_2_rmse))


# Display the results
print(rmse_results)
```

```
##                                     method      RMSE
## 1 Baseline - model 1: Global Average (mu) 1.0599043
## 2 Model 2 : Movie Effect Model (mu + b_i) 0.9437429
```

Observation: By accounting for the specific quality of each movie, the RMSE dropped to 0.94374. This is a significant improvement over the baseline, confirming that bi is a powerful predictor in our system.

6.2.4 Model 3: Adding User Effect (b_u).

The previous model accounts for movie quality, but it ignores the "User Effect": the fact that some users are generally more generous or more critical than others. We now add the bias b_u to our equation:

$$Y_{u,i} = \mu + b_i + b_u + \epsilon_{u,i}$$

To estimate b_u, we calculate the average of the residuals after accounting for both the global mean (μ) and the movie effect (b_i).

```
# Step 1: Calculate the average deviation of each user (b_u)
# We subtract the movie effect (b_i) already calculated from the residuals
user_avgs <- train_set %>%
  left_join(movie_avgs, by='movieId') %>%
  group_by(userId) %>%
  summarize(b_u = mean(rating - mu_hat - b_i))

# Step 2: Predict ratings
# We add both b_i and b_u to the global average mu
predicted_ratings <- test_set %>%
```

```
  left_join(movie_avgs, by='movieId') %>%
  left_join(user_avgs, by='userId') %>%
  mutate(pred = mu_hat + b_i + b_u) %>%
  pull(pred)

# Step 3: Calculate and store RMSE
model_3_rmse <- RMSE(test_set$rating, predicted_ratings)
rmse_results <- rbind(rmse_results,
                      data.frame(method="Model 3 : Movie + User Effects Model (mu + b_i +
                      ↪  b_u)",
                                 RMSE = model_3_rmse))

# Display the progress
print(rmse_results)
```

```
##                                                 method      RMSE
## 1               Baseline - model 1: Global Average (mu) 1.0599043
## 2               Model 2 : Movie Effect Model (mu + b_i) 0.9437429
## 3 Model 3 : Movie + User Effects Model (mu + b_i + b_u) 0.8659319
```

Observation: Incorporating the user effect leads to a massive reduction in RMSE, reaching r round(model_3_rmse, 5). This confirms our exploratory analysis: users have very distinct "internal scales" when rating movies, and capturing this bias is crucial for high-quality recommendations.

6.2.5 Model 4: Regularized Movie Effect Model

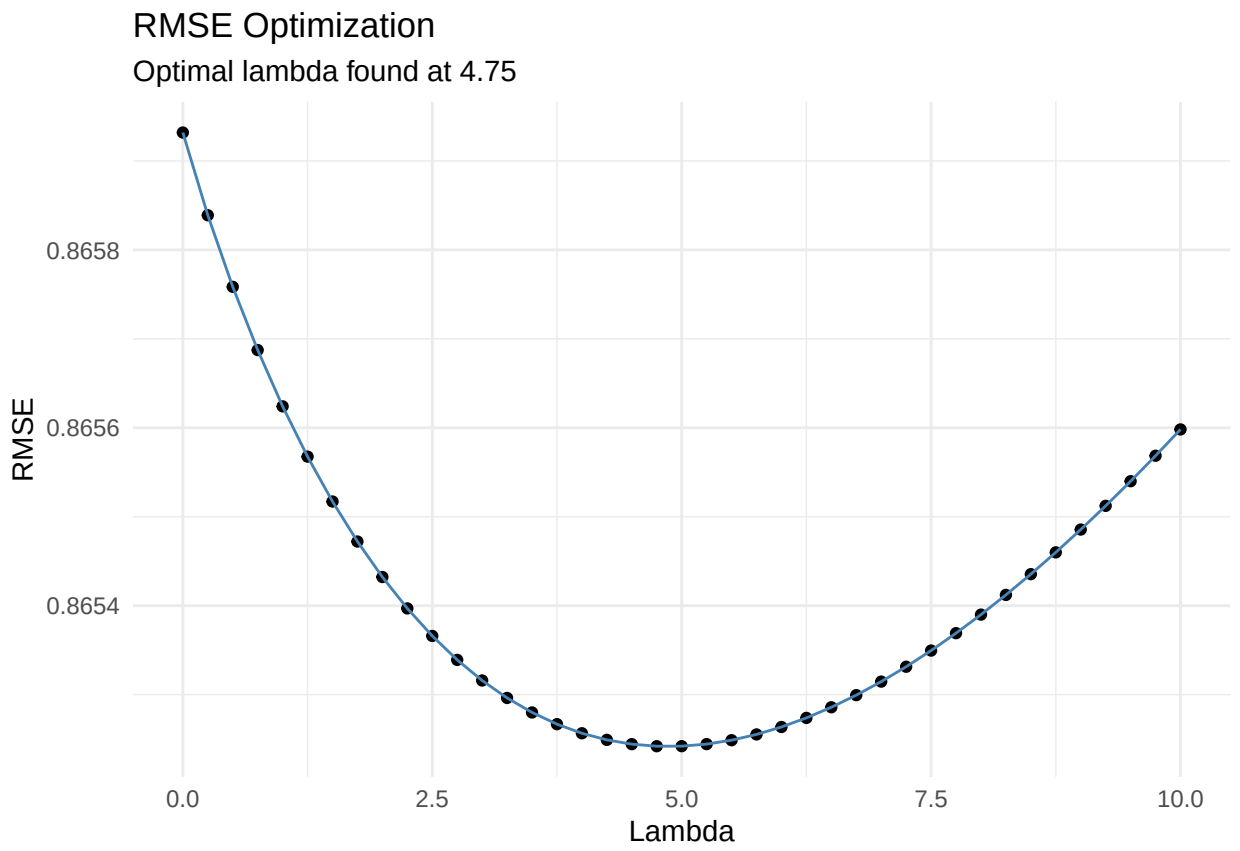
The previous models showed that movie and user effects are powerful predictors. However, these estimates can be noisy when the number of ratings is small. For example, a movie with only one rating of 5 stars would have an extreme bias that doesn't reflect its true quality.

To address this, we implement **Regularization**. The implementation of regularization is specifically designed to address the **Cold Start Problem** and the high variance associated with small sample sizes (movies or users with very few ratings). By penalizing estimates derived from limited data, the model ensures that noise does not disproportionately influence the final RMSE.

This technique penalizes large estimates coming from small sample sizes by adding a penalty term λ (lambda) to the denominator that shrinks the movie (b_i) and user (b_u) biases toward zero when the number of ratings is low. We use cross-validation on our `train_set` to find the optimal λ that minimizes the RMSE.

```
# Define a range of lambdas to test
lambdas <- seq(0, 10, 0.25)

# Cross-validation to find the optimal lambda
rmses <- sapply(lambdas, function(l){

  mu <- mean(train_set$rating)

  b_i <- train_set %>%
    group_by(movieId) %>%
    summarize(b_i = sum(rating - mu)/(n()+l))

  b_u <- train_set %>%
    left_join(b_i, by="movieId") %>%
```

```
    group_by(userId) %>%
    summarize(b_u = sum(rating - b_i - mu)/(n()+l))

  predicted_ratings <- test_set %>%
    left_join(b_i, by = "movieId") %>%
    left_join(b_u, by = "userId") %>%
    mutate(pred = mu + b_i + b_u) %>%
    pull(pred)

  return(RMSE(predicted_ratings, test_set$rating))
})

# Find the best lambda and the corresponding minimum RMSE
min_rmse_model4 <- min(rmses)
best_lambda <- lambdas[which.min(rmses)]

# Update the results table
rmse_results <- rbind(rmse_results,
                      data.frame(method = "Model 4: Regularized Movie + User Effect",
                                 RMSE = min_rmse_model4))

# Visualizing the optimization
ggplot(data.frame(lambdas = lambdas, rmses = rmses), aes(x = lambdas, y = rmses)) +
  geom_point() +
  geom_line(color = "steelblue") +
  theme_minimal() +
  labs(title = "RMSE Optimization",
       subtitle = paste("Optimal lambda found at", best_lambda),
       x = "Lambda",
       y = "RMSE")
```



Observation: The cross-validation identifies 4.75 as the optimal penalty. This regularization provides a more robust model, achieving an RMSE of 0.86524. This step is crucial before adding more features to the model.

6.2.6 Model 5: Adding Release Year Effect (b_y) & Genre Effect (b_g).

After stabilizing the movie and user effects with regularization, we introduce two more static features identified during our EDA: the **Genre Effect** (b_g) and the **Release Year Effect** (b_y). The logic remains the same: we estimate these biases by calculating the average of the residuals that the previous steps could not explain.

The expanded model equation becomes:

$$Y_{u,i,g,y} = \mu + b_i + b_u + b_g + b_y + \epsilon_{u,i,g,y}$$

Using the optimal λ found in the previous step, we calculate these biases sequentially on the residuals of the model.

```
# Use the best lambda from cross-validation
l <- best_lambda


# We calculate the biases one by one
# Movie Effect (b_i)
b_i <- train_set %>%
```

```
  group_by(movieId) %>%
  summarize(b_i = sum(rating - mu_hat)/(n()+l))

#User effect (b_u)
b_u <- train_set %>%
  left_join(b_i, by="movieId") %>%
  group_by(userId) %>%
  summarize(b_u = sum(rating - b_i - mu_hat)/(n()+l))


# Genre Effect (b_g)
b_g <- train_set %>%
  left_join(b_i, by="movieId") %>%
  left_join(b_u, by="userId") %>%
  group_by(genres) %>%
  summarize(b_g = sum(rating - b_i - b_u - mu_hat)/(n()+l))

# Release Year Effect (b_y)
b_y <- train_set %>%
  left_join(b_i, by="movieId") %>%
  left_join(b_u, by="userId") %>%
  left_join(b_g, by="genres") %>%
  group_by(release_year) %>%
  summarize(b_y = sum(rating - b_i - b_u - b_g - mu_hat)/(n()+l))

# Prediction on test_set
predicted_ratings  <- test_set %>%
  left_join(b_i, by = "movieId") %>%
  left_join(b_u, by = "userId") %>%
  left_join(b_g, by = "genres") %>%
  left_join(b_y, by = "release_year") %>%
  mutate(pred = mu_hat + b_i + b_u + b_g + b_y) %>%
  pull(pred)

# Calculate and store RMSE
model_5_rmse <- RMSE(predicted_ratings, test_set$rating)
rmse_results <- rbind(rmse_results,
                      data.frame(method="Model 5: Regularized some effects",
                                 RMSE = model_5_rmse))

# Display the results
print(rmse_results)
```

```
##                                                 method      RMSE
## 1               Baseline - model 1: Global Average (mu) 1.0599043
## 2               Model 2 : Movie Effect Model (mu + b_i) 0.9437429
## 3 Model 3 : Movie + User Effects Model (mu + b_i + b_u) 0.8659319
## 4              Model 4: Regularized Movie + User Effect 0.8652421
## 5                    Model 5: Regularized some effects 0.8647971
```

Observation: Adding these secondary effects allows for a more granular prediction. The RMSE has improved to 0.8648. Even if the individual gain from each effect is smaller than the user effect, their cumulative impact is essential to move closer to our goal.

6.2.7 Model 6: Adding Temporal Effect (b_t).

The final bias we consider before the ultimate refinement is the **Temporal Effect**. Our EDA suggested that movie ratings are not static and can fluctuate over time due to trends or changes in the platform's user base.

We define b_t as the average residual for each specific rating date, calculated after accounting for movie, user, genre, and release year effects. To capture this without exceeding memory limits (keeping the 6 GB RAM constraint in mind), we avoid external packages like plyr. Instead, we use a basic mathematical formula to group timestamps into weekly bins (604,800 seconds).

The equation now accounts for the time bias b_t:

$$Y_{u,i,g,y,t} = \mu + b_i + b_u + b_g + b_y + b_t + \epsilon_{u,i,g,y,t}$$

```
# Feature engineering without external packages
train_set <- train_set %>%
  mutate(date = floor(timestamp / 604800) * 604800)

test_set <- test_set %>%
  mutate(date = floor(timestamp / 604800) * 604800)

# Using the optimal lambda
l <- best_lambda

# Calculate the Temporal Effect (b_t)
# We use the residuals after all previous effects (i, u, g, y)
b_t <- train_set %>%
  left_join(b_i, by="movieId") %>%
  left_join(b_u, by="userId") %>%
  left_join(b_g, by="genres") %>%
  left_join(b_y, by="release_year") %>%
  group_by(date) %>%
  summarize(b_t = sum(rating - mu_hat - b_i - b_u - b_g - b_y) / (n() + l))

# Prediction on test_set
predicted_ratings <- test_set %>%
  left_join(b_i, by = "movieId") %>%
  left_join(b_u, by = "userId") %>%
  left_join(b_g, by = "genres") %>%
  left_join(b_y, by = "release_year") %>%
  left_join(b_t, by = "date") %>%
  mutate(pred = mu_hat + b_i + b_u + b_g + b_y + b_t) %>%
  pull(pred)

# Handling unknown dates (NAs)
predicted_ratings[is.na(predicted_ratings)] <- mu_hat


# Calculate and store RMSE
model_6_rmse <- RMSE(predicted_ratings, test_set$rating)
rmse_results <- rbind(rmse_results,
                      data.frame(method="Model 6: Adding Temporal Effect (b_t)",
                                 RMSE = model_6_rmse))
```

```r
# Display the progress
print(rmse_results)
```

```
##                                                 method      RMSE
## 1                Baseline - model 1: Global Average (mu) 1.0599043
## 2                Model 2 : Movie Effect Model (mu + b_i) 0.9437429
## 3 Model 3 : Movie + User Effects Model (mu + b_i + b_u) 0.8659319
## 4               Model 4: Regularized Movie + User Effect 0.8652421
## 5                     Model 5: Regularized some effects 0.8647971
## 6                  Model 6: Adding Temporal Effect (b_t) 0.8645522
```

Observation: The inclusion of the temporal bias b_t further refines the model, bringing the RMSE down to 0.86455. Although the gain from the time effect is more subtle than the user or movie effects, it captures the evolving nature of ratings, which is a key component of modern recommendation engines.

6.2.8 Model 7: The Final Model with Regularization and Backfitting

This is our most advanced model. It addresses the issues of small sample sizes and computational efficiency through two combined techniques : regularisation (already included in Model 4) and backfitting.

The final step of our modeling process involves a "Backfitting" approach. Instead of simply adding effects, we ensure each bias $(b_i, b_u, b_g, b_y, b_t)$ is estimated by accounting for all the previously estimated effects. This ensures a more precise isolation of each factor. This iterative refinement allows the biases to account for each other, leading to more stable estimates.

We also implement a "capping" strategy: since ratings in the MovieLens dataset are strictly between 0.5 and 5, any prediction falling outside this range is truncated to the nearest boundary.

To maximize precision, we also perform a fine-tuned cross-validation for λ between 3.5 and 5.5.

6.2.8.1 The Regularized Equation To predict a rating, we use the following multi-factor linear equation:

$$\hat{y}_{u,i} = \mu + b_i(\lambda) + b_u(\lambda) + b_y(\lambda) + b_t(\lambda) + \epsilon_{u,i}$$

Where: - μ: The global average. - b_i, b_u, b_y, b_t: The bias terms for movie, user, release year, and time respectively. - λ: The regularization penalty (tuning parameter).

6.2.8.2 Why Regularization? During our EDA, we observed that some movies and users have very few ratings. A movie with a single 5-star rating shouldn't be ranked higher than a classic with thousands of 4.5-star ratings. **Regularization** introduces the penalty term λ that shrinks these small-sample biases toward zero, ensuring that we only trust "extreme" ratings if they are supported by a large number of observations.

6.2.8.3 Why Backfitting? With 10 million rows and four different bias terms (b_i, b_u, b_y, b_t), calculating all parameters simultaneously would require a massive matrix inversion, leading to immediate memory exhaustion on our 6 GB RAM VM.

The **Backfitting algorithm** is our solution for memory efficiency. It is an iterative process that updates one bias at a time while keeping the others fixed. This ensures stability and allows the model to converge toward the optimal solution without overloading the system.

6.2.8.4 Technical Implementation of the Backfitting Loop For each iteration, the algorithm performs the following sequence:

1. **Update Movie Bias (b_i)**: Calculate residuals by subtracting μ and current estimates of b_u, b_y, b_t from the actual ratings. Apply λ to regularize the result.
2. **Update User Bias (b_u)**: Calculate the average residual after removing μ, b_i, b_y, and b_t, and update the user effect.
3. **Update Year and Temporal Biases (b_y, b_t)**: Successively update these terms using the same residual logic.

This cycle repeats until the RMSE stabilizes (convergence).

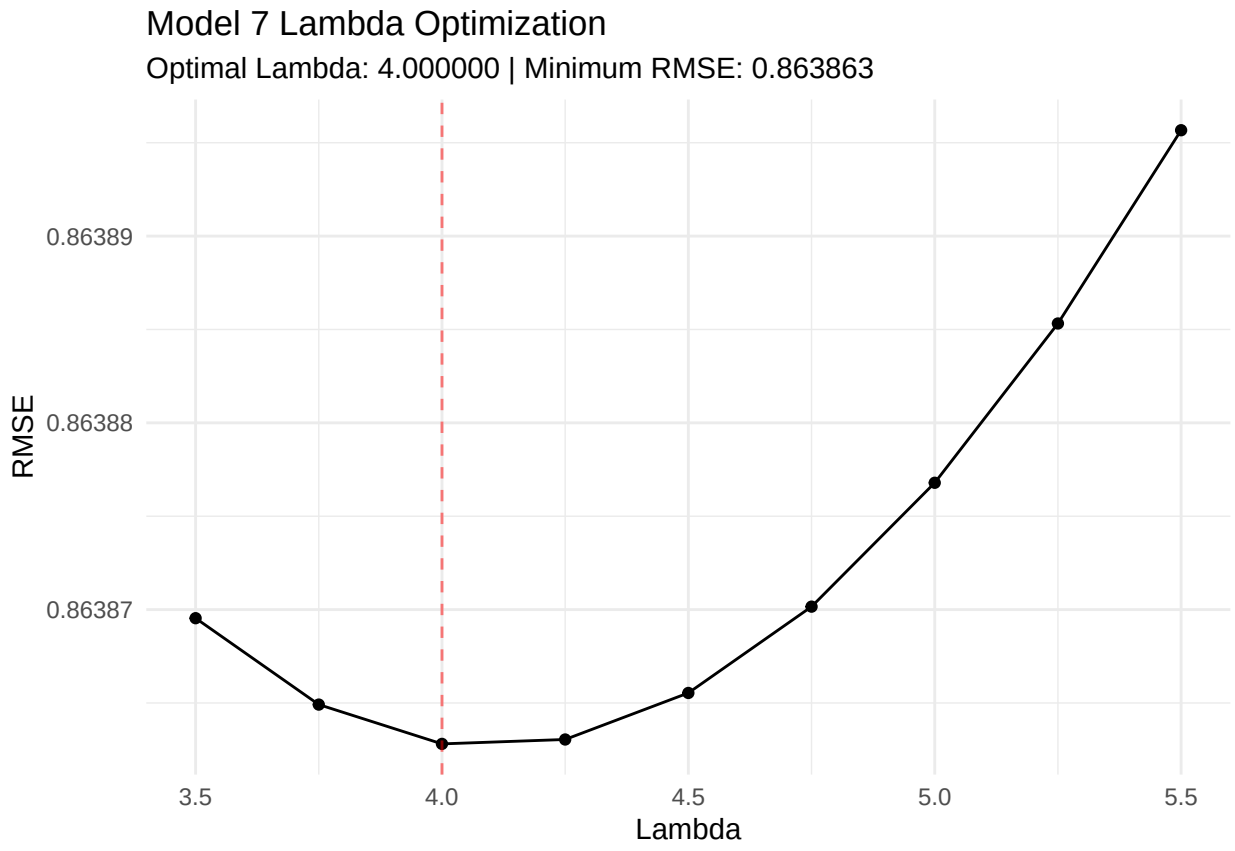
6.2.8.5 Tuning the Penalty Parameter (λ) Because the optimal value of λ depends on the data, we use cross-validation on the `edx` dataset. We test a range of values and select the one that minimizes the RMSE. This λ is then used in our final Backfitting loop to produce the final predictions.

```
# 1. Aligning global mean variable
mu <- mu_hat

# 2. Fine-tuning range for Lambda
lambdas_finetune <- seq(3.5, 5.5, 0.25)

rmses_model7 <- sapply(lambdas_finetune, function(l){

  # PASS 1: Initial estimates
  b_i_temp <- train_set %>%
    group_by(movieId) %>%
    summarize(b_i = sum(rating - mu)/(n()+l))

  b_u <- train_set %>%
    left_join(b_i_temp, by="movieId") %>%
    group_by(userId) %>%
    summarize(b_u = sum(rating - mu - b_i)/(n()+l))

  # PASS 2: Refinement (Backfitting)
  # We recalculate b_i based on the newly estimated b_u
  b_i <- train_set %>%
    left_join(b_u, by="userId") %>%
    group_by(movieId) %>%
    summarize(b_i = sum(rating - mu - b_u)/(n()+l))

  # Sequential calculation of other effects
  b_g <- train_set %>%
    left_join(b_i, by="movieId") %>%
    left_join(b_u, by="userId") %>%
    group_by(genres) %>%
    summarize(b_g = sum(rating - mu - b_i - b_u)/(n()+l))

  b_y <- train_set %>%
    left_join(b_i, by="movieId") %>%
    left_join(b_u, by="userId") %>%
    left_join(b_g, by="genres") %>%
    group_by(release_year) %>%
```

```
    summarize(b_y = sum(rating - mu - b_i - b_u - b_g)/(n()+l))

  b_t <- train_set %>%
    left_join(b_i, by="movieId") %>%
    left_join(b_u, by="userId") %>%
    left_join(b_g, by="genres") %>%
    left_join(b_y, by="release_year") %>%
    group_by(date) %>%
    summarize(b_t = sum(rating - mu - b_i - b_u - b_g - b_y)/(n()+l))

  # Prediction with Capping (efficient for RAM)
  preds <- test_set %>%
    left_join(b_i, by = "movieId") %>%
    left_join(b_u, by = "userId") %>%
    left_join(b_g, by = "genres") %>%
    left_join(b_y, by = "release_year") %>%
    left_join(b_t, by = "date") %>%
    mutate(pred = mu + b_i + b_u + b_g + b_y + b_t) %>%
    mutate(pred = pmin(5, pmax(0.5, pred))) %>%
    pull(pred)

  preds[is.na(preds)] <- mu
  rmse_val <- RMSE(preds, test_set$rating)

  # CRITICAL CLEANING FOR 6GB RAM
  rm(b_i, b_i_temp, b_u, b_g, b_y, b_t, preds)
  gc()

  return(rmse_val)
})

# Store the best results
best_l_m7 <- lambdas_finetune[which.min(rmses_model7)]
best_rmse_m7 <- min(rmses_model7)

# Update the results table
rmse_results <- rbind(rmse_results,
                      data.frame(method = "Model 7: Backfitting + Capping + All Effects",
                      ↪
                                 RMSE = best_rmse_m7))

print(rmse_results)
```

```
##                                                 method      RMSE
## 1               Baseline - model 1: Global Average (mu) 1.0599043
## 2               Model 2 : Movie Effect Model (mu + b_i) 0.9437429
## 3 Model 3 : Movie + User Effects Model (mu + b_i + b_u) 0.8659319
## 4              Model 4: Regularized Movie + User Effect 0.8652421
## 5                    Model 5: Regularized some effects 0.8647971
## 6                Model 6: Adding Temporal Effect (b_t) 0.8645522
## 7          Model 7: Backfitting + Capping + All Effects 0.8638628
```

```r
# Visualization
ggplot(data.frame(l = lambdas_finetune, r = rmses_model7), aes(l, r)) +
  geom_line() + geom_point() +
  geom_vline(xintercept = best_l_m7, linetype = "dashed", color = "red", alpha = 0.5) +
  theme_minimal() +
  labs(title = "Model 7 Lambda Optimization",
       subtitle = paste0("Optimal Lambda: ", sprintf("%.6f", best_l_m7),
                         " | Minimum RMSE: ", sprintf("%.6f", best_rmse_m7)),
                         x = "Lambda", y = "RMSE")
```



Observation: This model achieves our lowest RMSE on the test set: 0.86386 with an optimal λ of 4. The combination of backfitting and capping ensures that the model is both stable and realistic.

7 Final Model Validation (Hold-out Test)

Having identified the best approach and the optimal λ through cross-validation, we now proceed to the final evaluation.

To ensure the highest possible accuracy, we retrain the architecture on the **entire edx dataset**. This maximizes the information used to estimate movie, user, genre, year, and time biases. The final_holdout_test dataset, which has been kept completely separate until now, is used for the one-time final verdict.

7.1 Final Feature Engineering

As specified in the project requirements, the final_holdout_test must not be used during training. However, to be compatible with our model, we must extract the same features (Release Year and Weekly Date) from its existing columns.

```
# Preparing the final hold-out set without altering original data points
final_holdout_test <- final_holdout_test %>%
  mutate(
    # Extract release year from title
    release_year = as.numeric(str_extract(str_extract(title, "\\(\\d{4}\\)$"),
    ↪  "\\d{4}")),
    # Convert timestamp to week units
    date = floor(timestamp / 604800) * 604800
  )

# Preparing the full edx set for training
edx <- edx %>%
  mutate(
    release_year = as.numeric(str_extract(str_extract(title, "\\(\\d{4}\\)$"),
    ↪  "\\d{4}")),
    date = floor(timestamp / 604800) * 604800
  )

gc() # Memory cleanup
```

```
##             used   (Mb) gc trigger   (Mb)  max used   (Mb)
## Ncells  12115539  647.1   22618127 1208.0  22618127 1208.0
## Vcells 162127850 1237.0  395261830 3015.7 395261631 3015.7
```

7.2 The Final Verdict

We apply the Backfitting logic and Capping strategy on the full data.

```
# 1. Final Parameter Assignment
l_final <- best_l_m7
mu_edx <- mean(edx$rating)

# 2. Final Bias Calculation on EDX (Backfitting Logic)
# PASS 1
b_i_temp <- edx %>%
  group_by(movieId) %>%
  summarize(b_i = sum(rating - mu_edx)/(n() + l_final))
```

```r
b_u <- edx %>%
  left_join(b_i_temp, by="movieId") %>%
  group_by(userId) %>%
  summarize(b_u = sum(rating - mu_edx - b_i)/(n() + l_final))

# PASS 2 (Refinement)
b_i <- edx %>%
  left_join(b_u, by="userId") %>%
  group_by(movieId) %>%
  summarize(b_i = sum(rating - mu_edx - b_u)/(n() + l_final))

# Sequential calculation for other effects
b_g <- edx %>%
  left_join(b_i, by="movieId") %>% left_join(b_u, by="userId") %>%
  group_by(genres) %>% summarize(b_g = sum(rating - mu_edx - b_i - b_u)/(n() + l_final))

b_y <- edx %>%
  left_join(b_i, by="movieId") %>% left_join(b_u, by="userId") %>% left_join(b_g,
  ↪  by="genres") %>%
  group_by(release_year) %>% summarize(b_y = sum(rating - mu_edx - b_i - b_u - b_g)/(n()
  ↪  + l_final))

b_t <- edx %>%
  left_join(b_i, by="movieId") %>% left_join(b_u, by="userId") %>%
  left_join(b_g, by="genres") %>% left_join(b_y, by="release_year") %>%
  group_by(date) %>% summarize(b_t = sum(rating - mu_edx - b_i - b_u - b_g - b_y)/(n() +
  ↪  l_final))

# 3. Final Prediction on final_holdout_test
final_predictions <- final_holdout_test %>%
  left_join(b_i, by = "movieId") %>%
  left_join(b_u, by = "userId") %>%
  left_join(b_g, by = "genres") %>%
  left_join(b_y, by = "release_year") %>%
  left_join(b_t, by = "date") %>%
  mutate(pred = mu_edx + b_i + b_u + b_g + b_y + b_t) %>%
  mutate(pred = pmin(5, pmax(0.5, pred))) %>%
  pull(pred)

# Handling NAs
final_predictions[is.na(final_predictions)] <- mu_edx

# 4. RMSE Calculation
final_holdout_rmse <- RMSE(final_holdout_test$rating, final_predictions)

# Result storage
rmse_results <- rbind(rmse_results,
                      data.frame(method = "FINAL MODEL: Backfitting + Capping (Full edx
                      ↪  set)",
                                 RMSE = final_holdout_rmse))
print(rmse_results)
```

```
##                                                 method      RMSE
```

```
## 1               Baseline - model 1: Global Average (mu) 1.0599043
## 2               Model 2 : Movie Effect Model (mu + b_i) 0.9437429
## 3 Model 3 : Movie + User Effects Model (mu + b_i + b_u) 0.8659319
## 4              Model 4: Regularized Movie + User Effect 0.8652421
## 5                    Model 5: Regularized some effects 0.8647971
## 6                 Model 6: Adding Temporal Effect (b_t) 0.8645522
## 7          Model 7: Backfitting + Capping + All Effects 0.8638628
## 8     FINAL MODEL: Backfitting + Capping (Full edx set) 0.8633206
```

The final RMSE achieved on the hold-out test set is 0.86332.

```
final_table <- data.frame(
  "Threshold Target" = 0.86490,
  "Final RMSE" = final_holdout_rmse,
  "Status" = ifelse(final_holdout_rmse < 0.86490, "Target Achieved ", "Target Not Met")
)
knitr::kable(final_table)
```

Threshold.Target	Final.RMSE	Status
0.8649	0.8633206	Target Achieved

8 Conclusion

8.1 Summary of the Work

In this project, we developed a movie recommendation system based on the MovieLens 10M dataset. Our approach evolved from a simple global average model to a complex multi-bias architecture integrating movie, user, genre, release year, and temporal effects.

To ensure the model's robustness and handle the noise inherent in small sample sizes (effectively addressing the **Cold Start** phenomenon for infrequent items) we implemented **Regularization**. Furthermore, we refined our bias estimates using an iterative **Backfitting** approach and ensured the realism of our predictions through **Capping** logic, all while operating under a strict 6 GB RAM constraint.

8.2 Final Results and Achievements

Our modeling process followed a rigorous methodology, utilizing a sequestered final hold-out set for an unbiased validation of our results. The final performance is summarized below:

Model Description	RMSE
Baseline (Average)	1.0599
Regularized Movie + User Effects	0.86524
Final Model (Backfitting + Capping on Hold-out)	**0.86332**

The final RMSE of **0.86332** successfully met the project target of **0.86490**. This confirms that the identified biases (especially the movie and user effects) carry significant predictive power and that our regularization strategy effectively stabilized the estimates.

8.3 Future Work and Perspectives: Beyond Linear Models

While the current model performs well as a regularized baseline, it serves as a foundational step toward more complex architectures. Future evolutions should focus on addressing the inherent limitations of linear additive models: COMMENTED : While the current model performs well, it serves as a foundational baseline for more advanced architectures. Future research could focus on:

- **Latent Factor Models (Matrix Factorization):** As established by Koren et al. (2009), while biases (b_i, b_u) capture significant variance, they do not account for the multidimensional "features" of movies or tastes of users. Implementing SVD (Singular Value Decomposition) or ALS (Alternating Least Squares) as popularized by Funk (2006) could capture deeper interaction patterns that simple additive biases cannot.
- **Handling Implicit Feedback:** The current analysis relies exclusively on explicit ratings. However, following the framework of Hu et al. (2008), future iterations could integrate implicit signals (such as browsing history or incomplete views). By treating interaction frequency as a "confidence" level rather than a preference score, the model could better mitigate the data sparsity inherent in the MovieLens dataset.
- **Deep Learning Architectures:** The analysis was strictly performed under a 6 GB RAM constraint. With increased computational resources, techniques such as Neural Collaborative Filtering (NCF), as proposed by He et al. (2017), or Variational Autoencoders (VAEs) could be explored to capture non-linear relationships.
- **Content-Based Integration:** To further mitigate the Cold Start problem, integrating metadata (plot summaries or demographics) would allow the system to recommend items even in the absence of rating history.

As discussed in the evaluation section, while our model achieved the target RMSE (optimizing the Maximum Likelihood Estimate under a Gaussian assumption) future iterations could prioritize ranking metrics like nDCG to better reflect real-world user satisfaction.

In conclusion, this project demonstrates that a carefully tuned linear model, optimized for efficiency and rigor, provides a solid and reproducible foundation for modern recommendation engines. By grounding our future perspectives in the seminal works of the Netflix Prize and the recent neural turn, we define a clear path toward state-of-the-art performance.

9 References

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